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Design Principles for Building High Performance Clusters

Networking Fundamentals

Ashrut Ambastha
NVIDIA



National
Supercom
puting
Mission



Centre for
Development
of
Advanced
Computing



Networks

Interlinking of Multiple Compute Elements

SMP?? Clusters??

Cache Coherency?? Message Passing??

Bandwidth?? Latency??

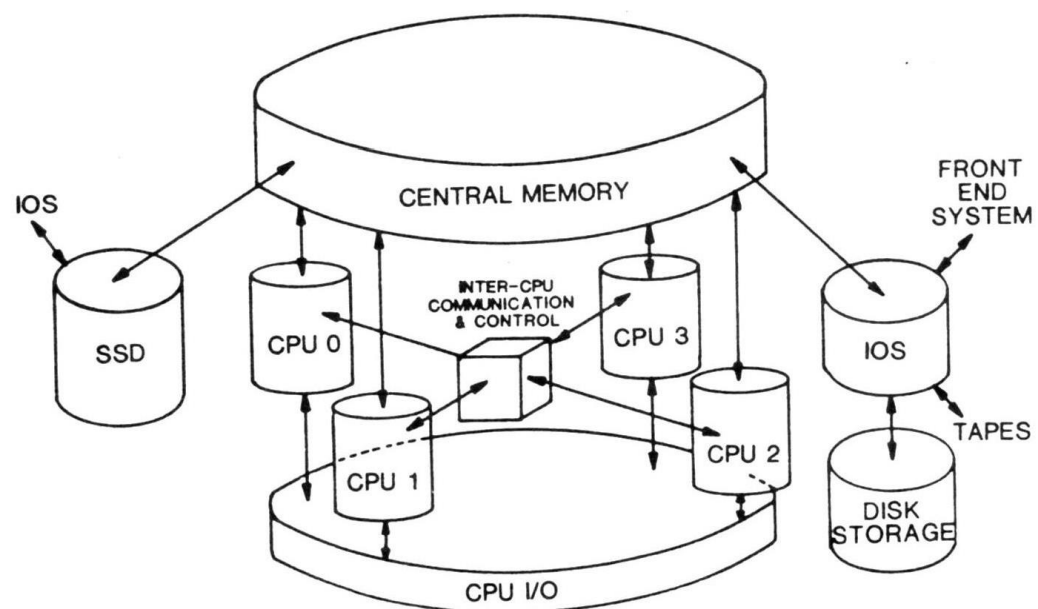




Traditional SMP Machines



*CRAY X-MP: <https://commons.wikimedia.org/wiki/File:BSC-CRAY-X-MP-EA-A.JPG>



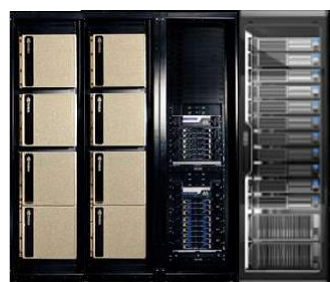


Modern Supercomputing Clusters





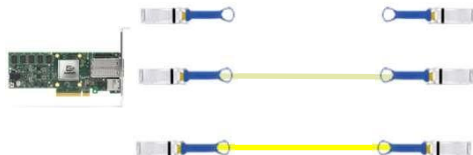
Network Components for High-Performance Clusters



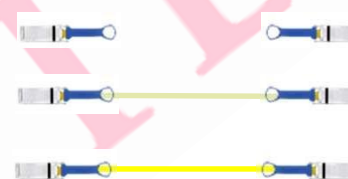
Compute
Servers
GPUs, CPUs



Optical
Cables & Transceivers
Ethernet
InfiniBand



Network
Switches



Storage
Subsystem
HDD/SSD/Flash



Network Switches, Network Adapters, Cables & Transceivers



HPC Interconnect History and Development



First Teraflop Supercomputer
Sandia ASCI Red
Intel



First Petaflop Supercomputer
LANL Roadrunner
IBM / Mellanox InfiniBand



QsNet



QsNet with Gateway to Ethernet

Myricom

Myrinet



InfiniPath
(InfiniBand)

TrueScale
(InfiniBand)

OmniPath

CRAY

Crossbar

Seastar

Gemini

Aries

Slingshot



InfiniBandSDR

DDR

QDR

FDR

EDR

HDR

NDR

XDR

1995

2000

2005

2010

2015

2020

2025



Fundamentals of the Physical Layer



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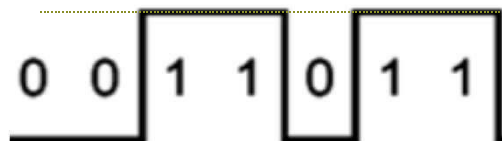


Transmitting Bits and Bytes

NRZ

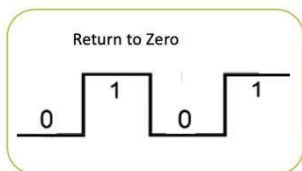
Non-Return to Zero

1G, 10G, 25G NRZ
(1 bit/clock)



NRZ allows for multiple
1's in a row without
having to return to zero

Return-to-Zero
After each data pulse



NRZ used in:
1G 1x1G
10G 1x10G
40G 4x10G

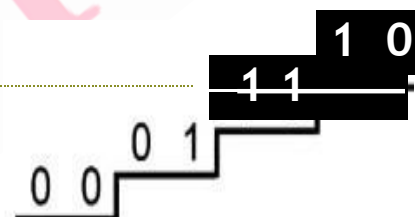
25G 1x25G
100G 4x25G

- + Doubles the data rate
- + Keeps same 25GHz clock
- + Enables lower cost material system

PAM4

Pulse Amplitude Modulation 4-levels

50G-PAM4 Signaling
(2 bits/clock)



PAM4 used in:
200G 4x50G-PAM4
400G 8x50G-PAM4

400G 4x100G-PAM4
800G 8x100G-PAM4



What We Don't Want to Achieve

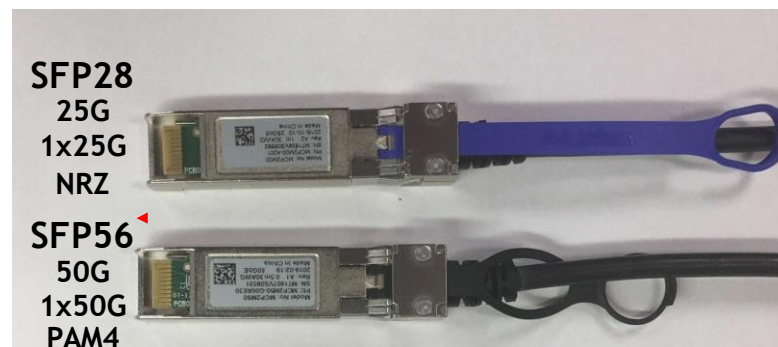
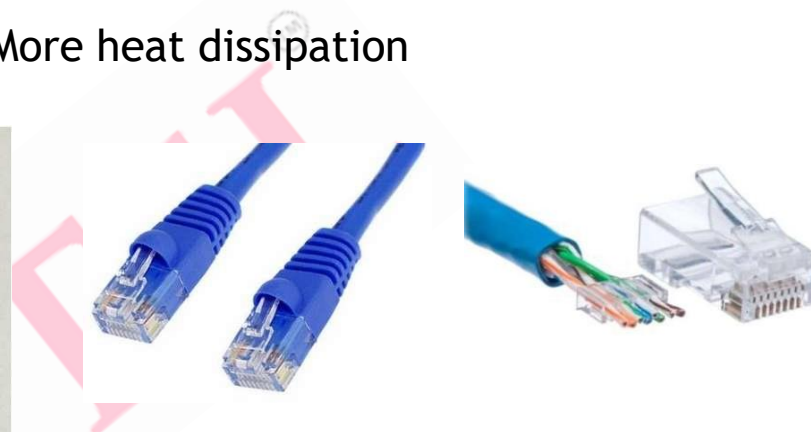
HOW STANDARDS PROLIFERATE:





Connector Evolution and Need

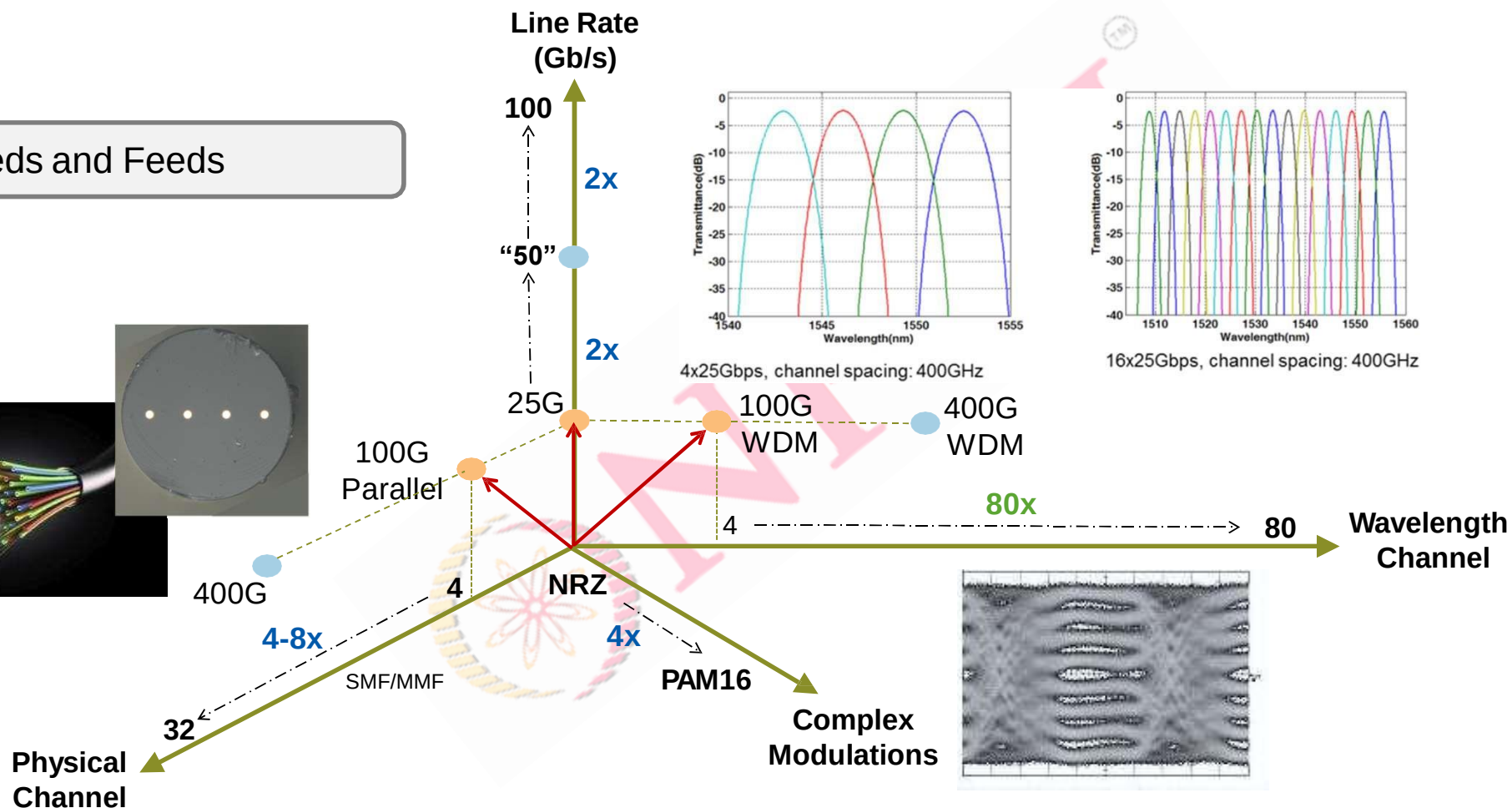
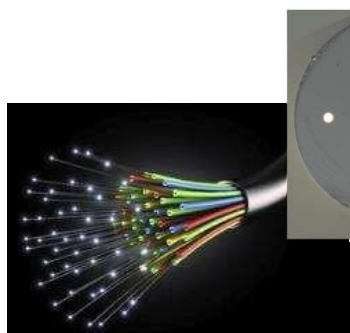
More space for optics & electronics; More heat dissipation





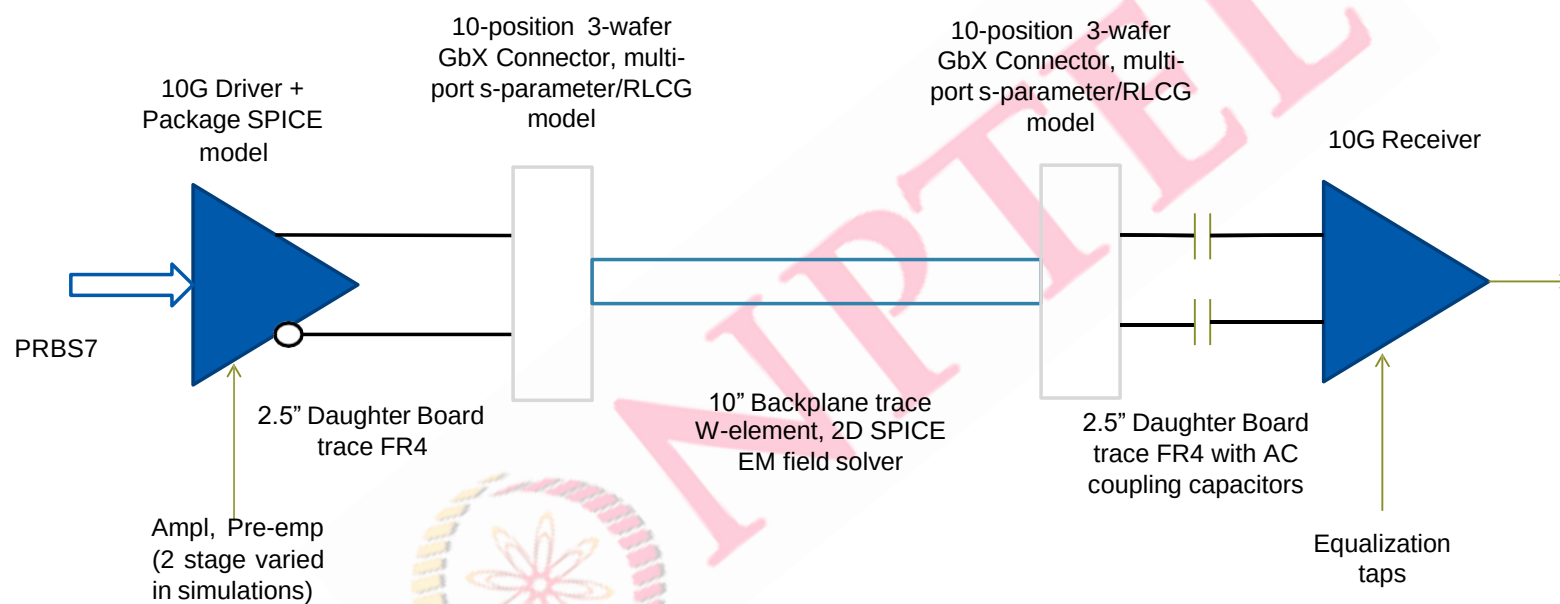
Interconnect Speeds and Modulation

Speeds and Feeds





An Example High-Speed Channel

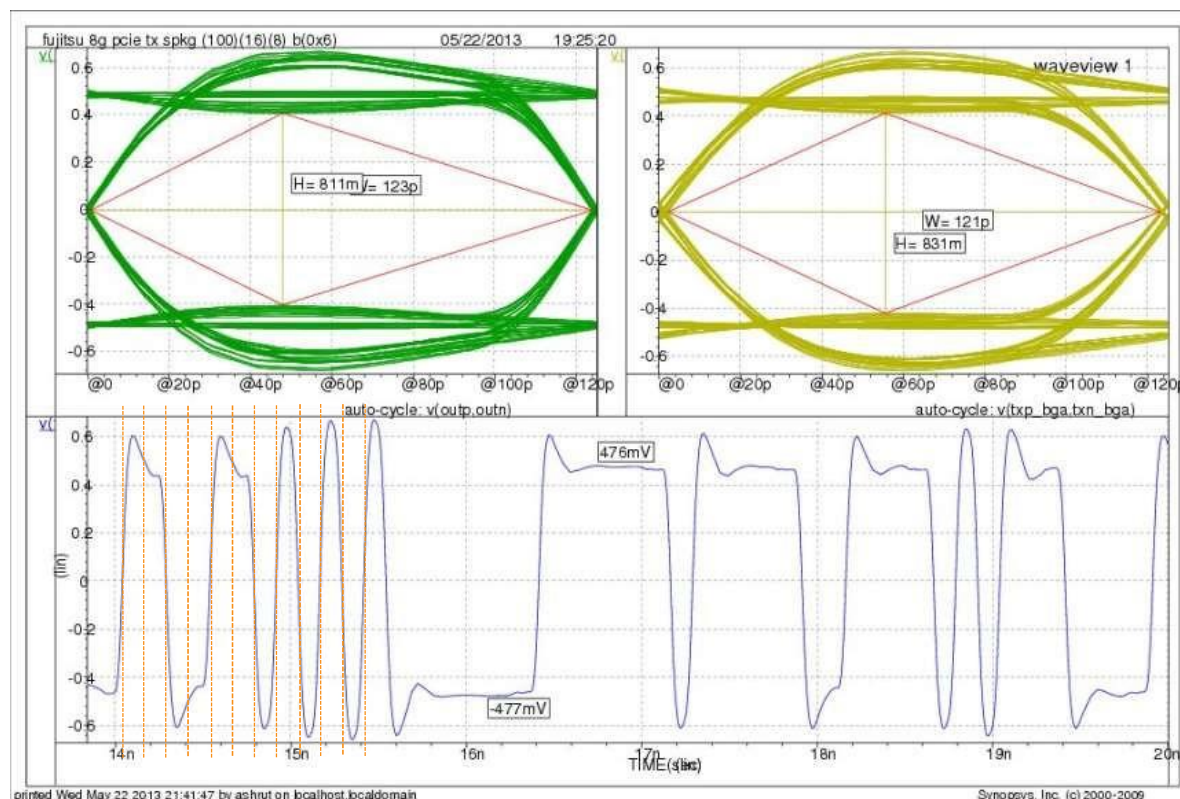




The Eye Diagram

- For high-speed signaling, the most fundamental unit of measure or visual-aid is the eye-diagram
- Nothing but a representation of 0's and 1's in a "folded" time space.
- X-axis represents 1-bit period
- Y-axis represents waveform amplitude
- But this does not look very much like an eye....

PCIe Gen3 eye-diagram at o/p buffer (before and after package model)



printed Wed May 22 2013 21:41:47 by ashut on localhost.localdomain

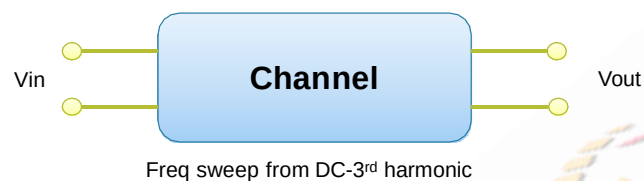
Synopsys, Inc. (c) 2000-2009



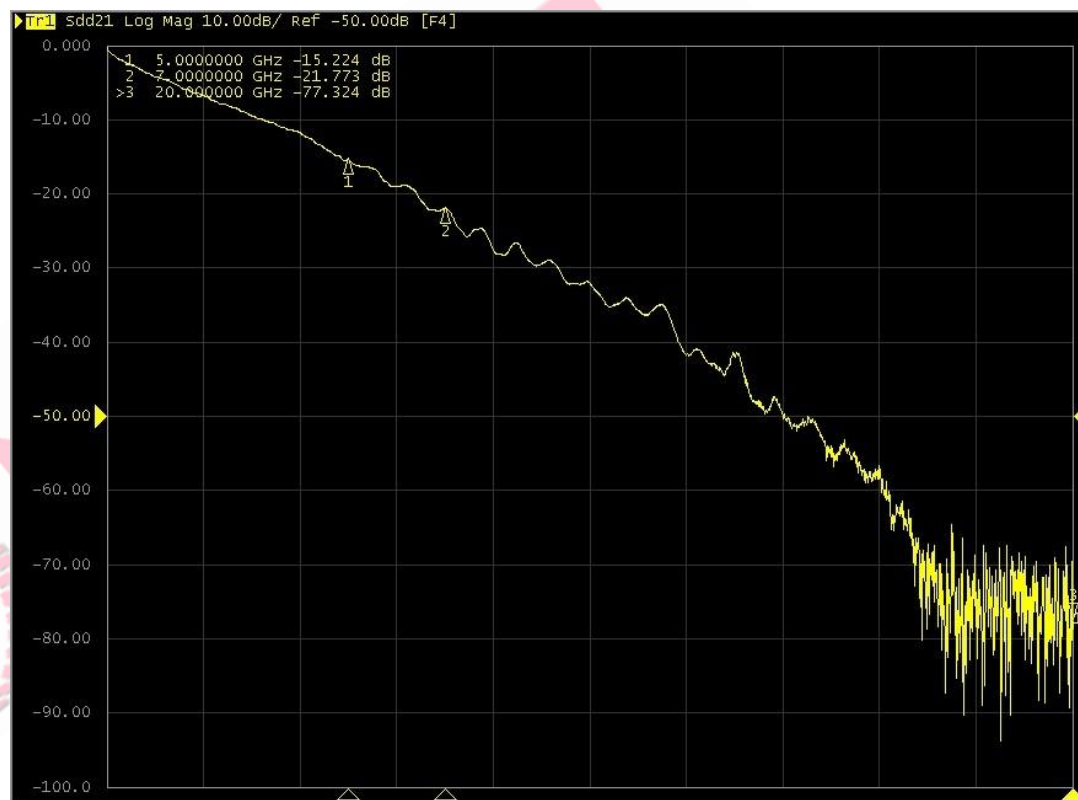
Insertion Loss or Sdd21

Without going into the incident/reflected power and scattering matrix of 2-port networks.

Insertion loss can be simply defined as $20 \cdot \log_{10}$ of “forward voltage attenuation”



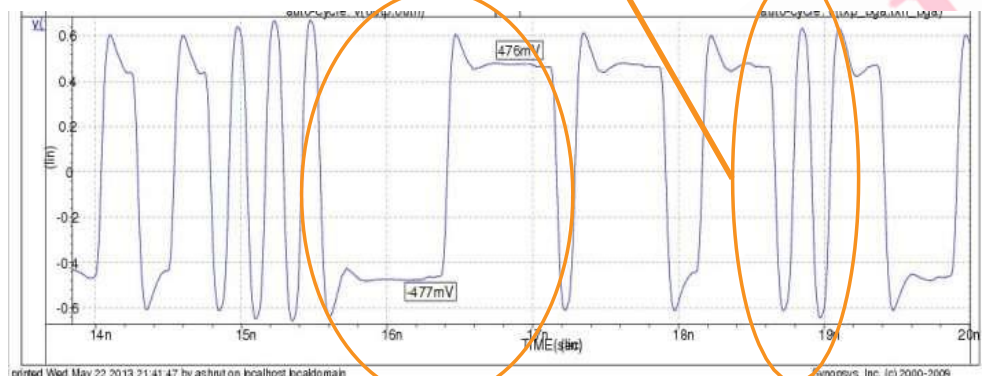
$$IL_{(dB)} = 20 \cdot \log(V_{out}/V_{in})$$





Insertion Loss or Sdd21 cont..

- IL starts increasing as frequency increases



Will face lesser attenuation

Will be attenuated more

- IL is simply related to copper trace length and channel dielectric material properties
- Longer the channel (trace) length higher the loss



Cross-section view of a Transmission line

- Dielectric's ability to polarize/de-polarize causes freq dependence of Insertion Loss



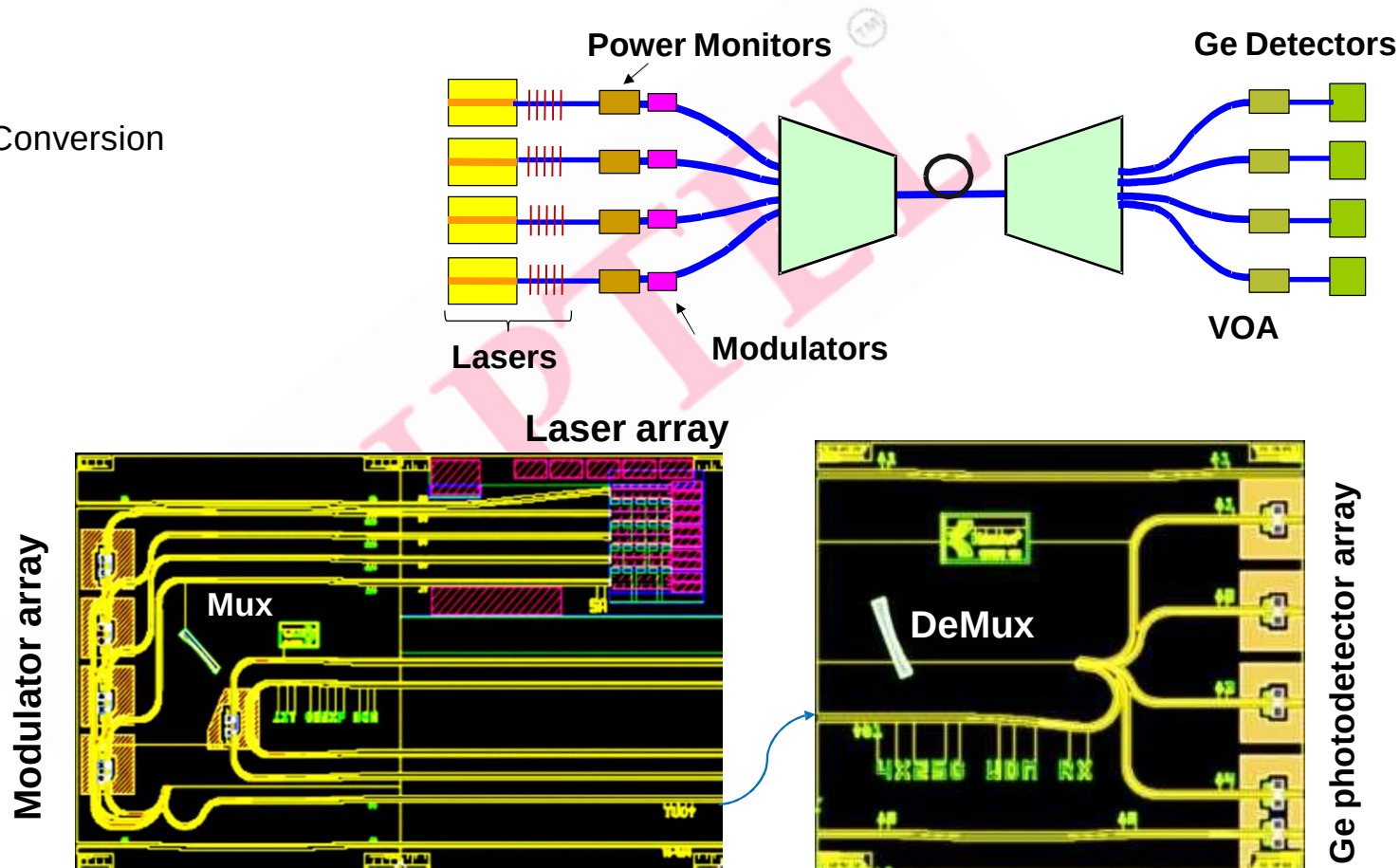
Enter Optics.. VCSEL or Photonics

Electrical-Optical-Electrical Conversion

- Lasers
- Modulators
- Detectors



QSFP





Network Layer and Routing





HDR InfiniBand Switch

QM8700, 1U Series

40 QSFP ports (PAM4, 50Gb/s per lane)

40 HDR (200Gb/s)

80 HDR100 (100Gb/s)

130nsec latency

390M messages per second (64Byte)

16Tb/s aggregated bandwidth

SHARPV2 low latency data reduction and streaming aggregation

22" depth

6 fans (5+1), hot swappable

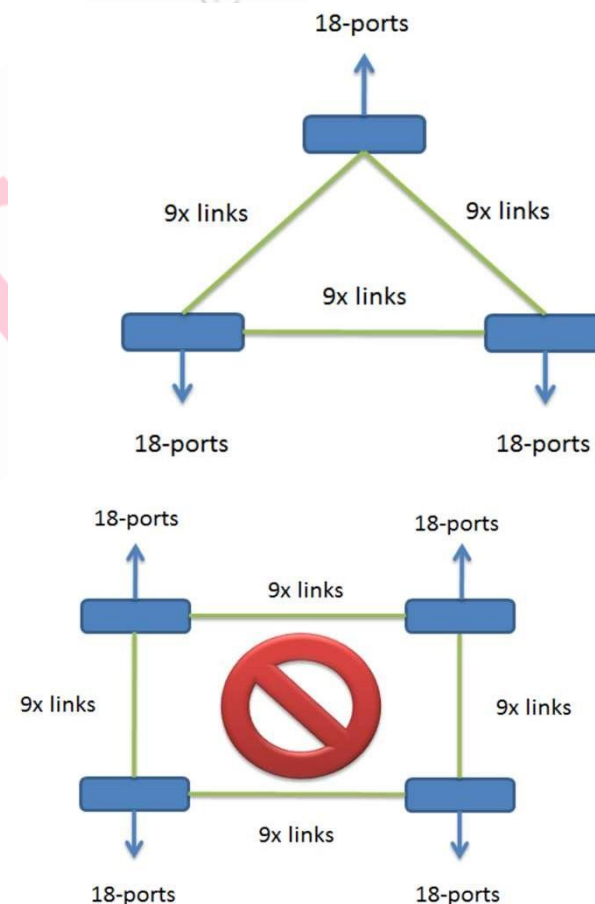
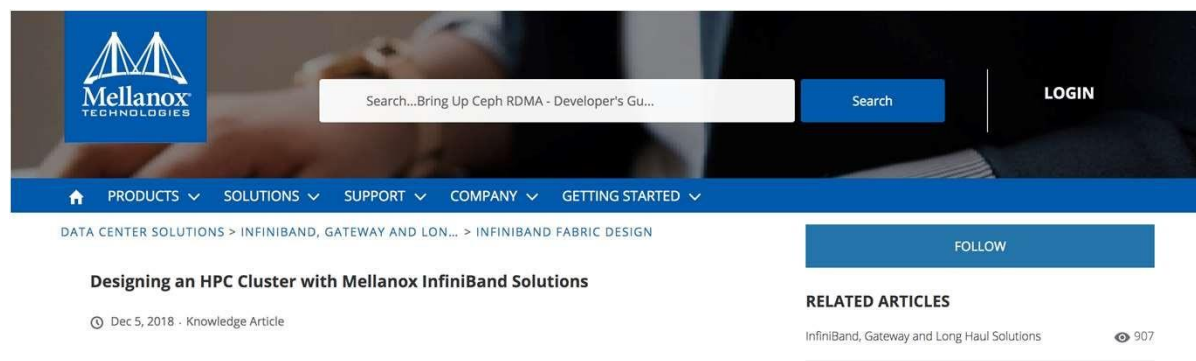
2 power supplies (1+1), hot swappable



<https://www.nvidia.com/en-in/networking/infiniband/qm8700/>



Designing Small HPC/AI Clusters

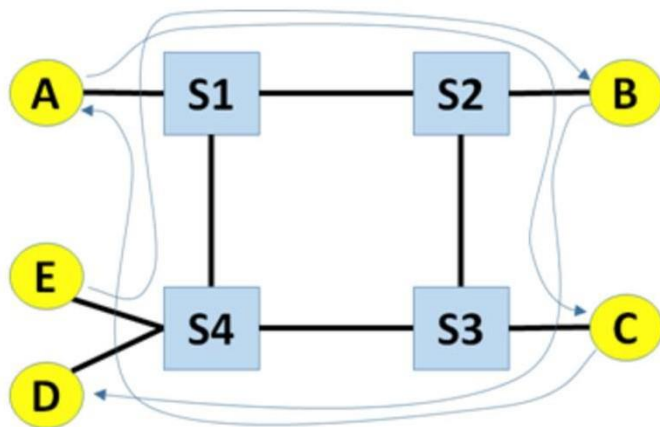




Credit Loops

Like many other networks, **InfiniBand dislikes loops**. Specifically, it dislikes **logical loops** where link back pressure can create a deadlock situation. These are called **credit loops**. Although the HOQ timers periodically clear such a deadlock, performance suffers. A credit loop only represents a potential deadlock, which depends on the traffic at each link in the loop. However, at InfiniBand speeds such a deadlock can occur very quickly given the right traffic pattern.

The following diagram illustrates a very simple credit loop.

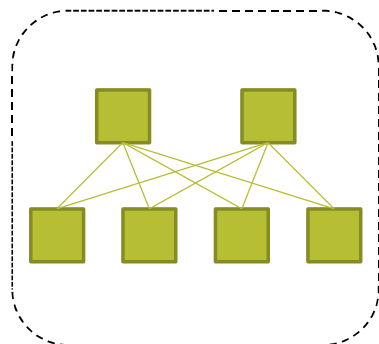


In this example:

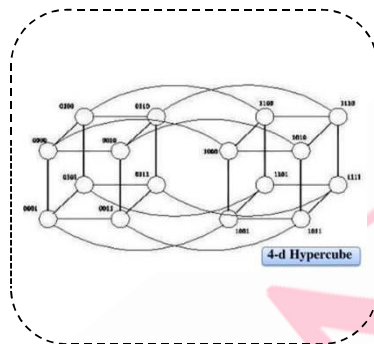
- Four Switch ASICs are connected in a ring topology.
- A host adapter (HCA) connects a server (Node) to each Switch, with an additional HCA on Switch 4.



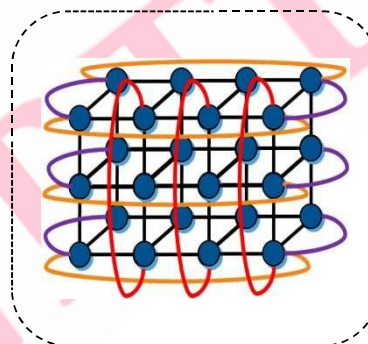
Network Topologies



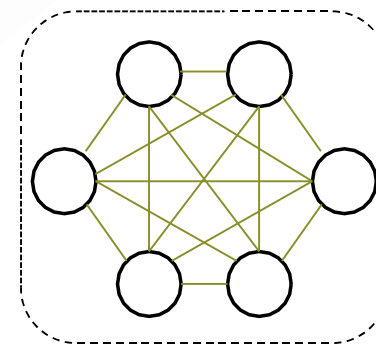
Fat Tree



Hypercube



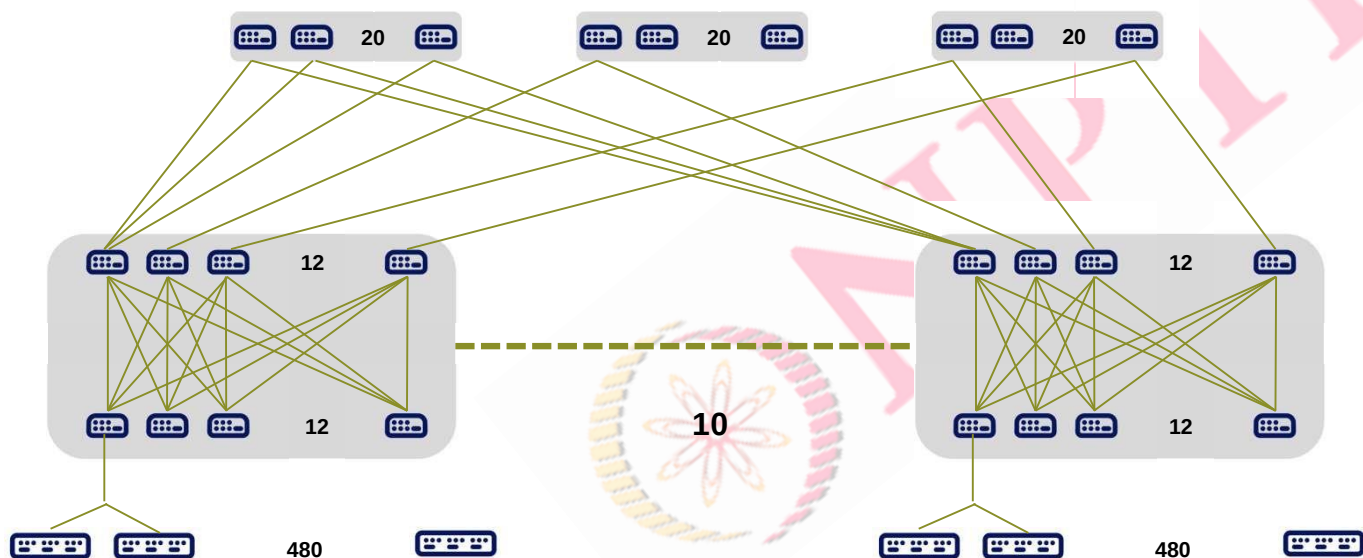
Torus



Dragonfly



3-level Clos with 1U HDR switches



■ 300 ASICs

■ $40 \times 12 = 480$ node sub-cluster (S1-S10)

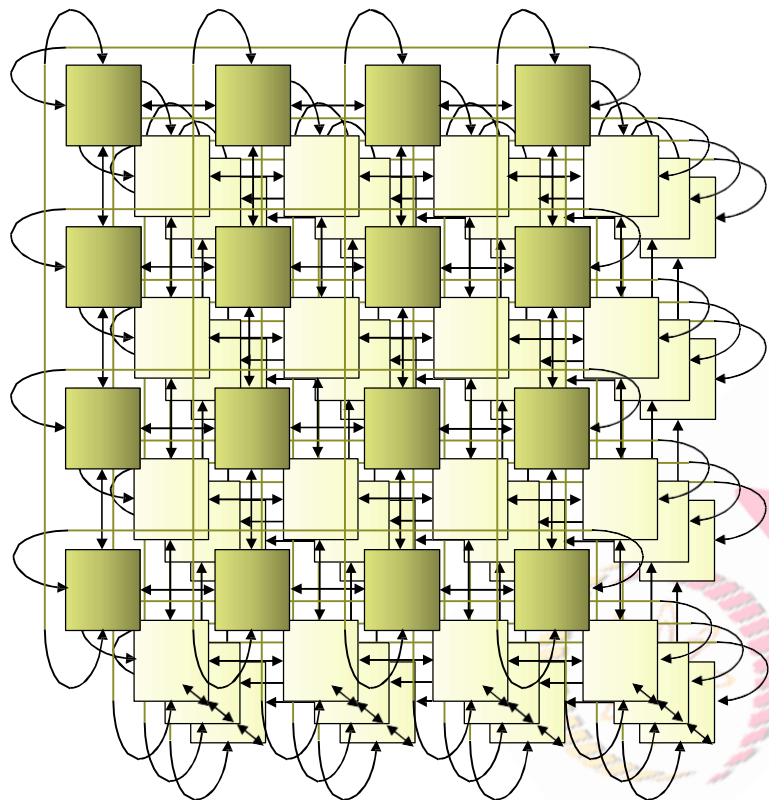
■ 10 Sub-clusters = 4800 port system

■ Expansion requires re-design

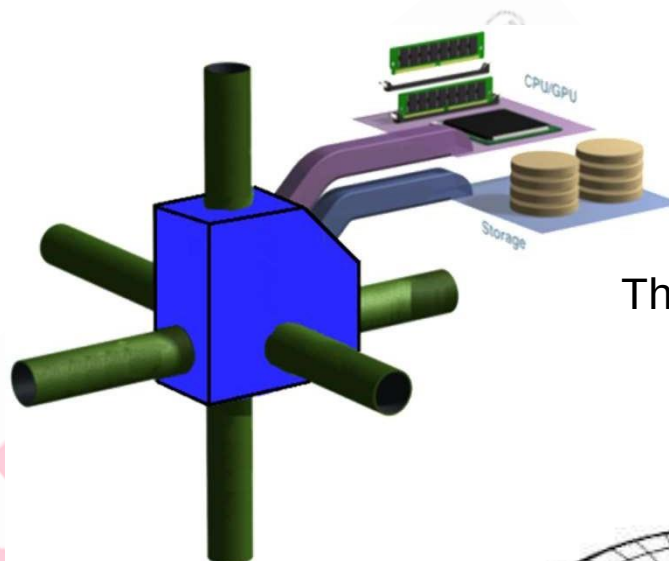
■ 5-hop network across any node-pair



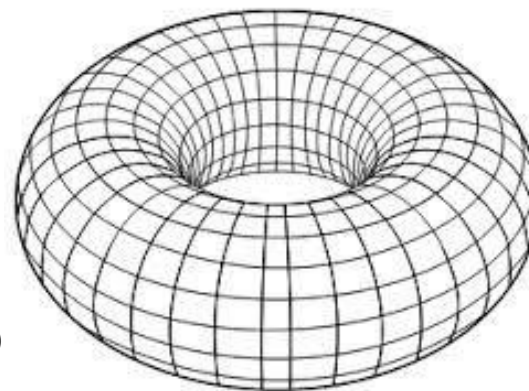
The 3-D Torus



A 4x4x4 Torus



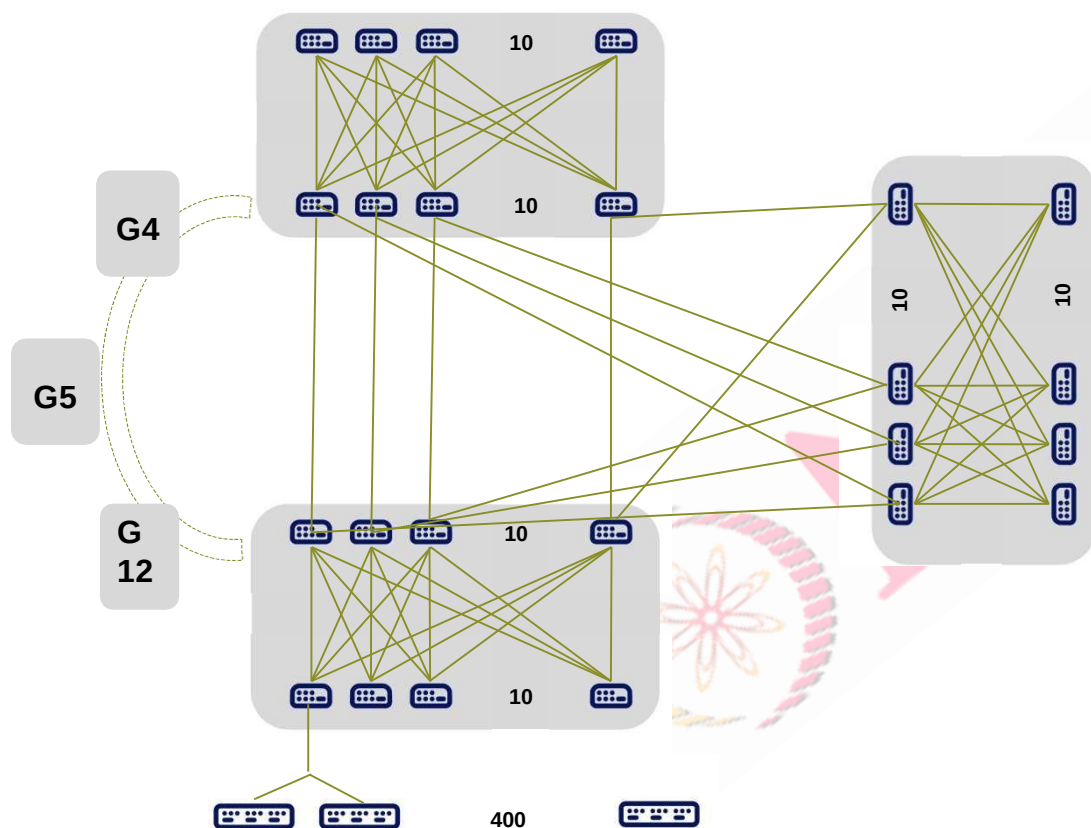
The Building Block



Natural "folding" to create a Torus



Dragonfy+ {240 ASICs}



- All-connect “Mesh” of “Groups”
- Each Group is full bi-partite
 - 20x 40p 200G switch per group
 - 200x 200G intra-group links per switch
 - 200x 200G inter-group links per switch
- 400 servers per group
- System can accommodate up-to 20 groups {8000 servers}
- Min-hop routing will have 4-hops across any node pair



Infiniband Packet Format

- Packets are routable end-to-end fabric unit of transfer
 - Link management packets: train and maintain link operation
 - Data packets
 - Send
 - Read
 - Write
 - Acks

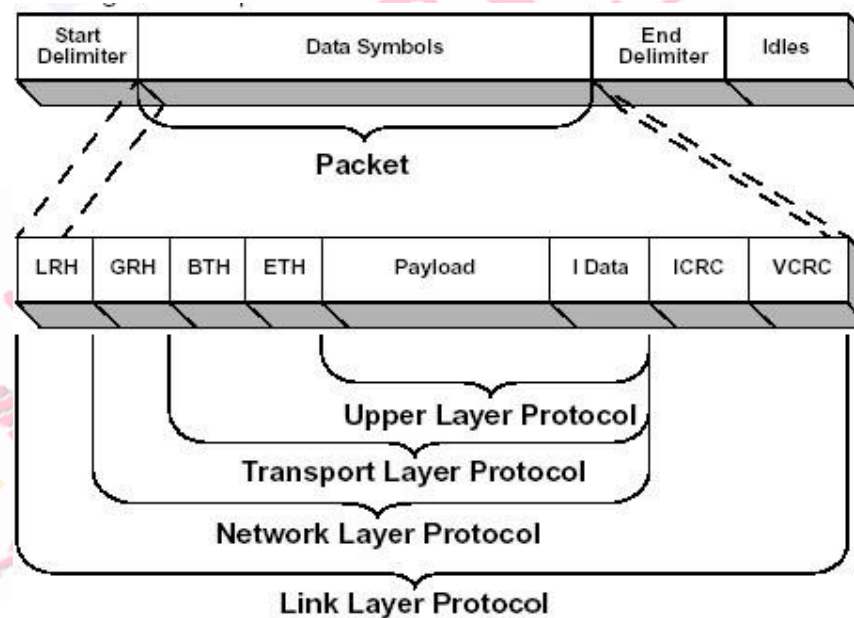


Figure 27 IBA Data Packet Format



Transport Layer and RDMA

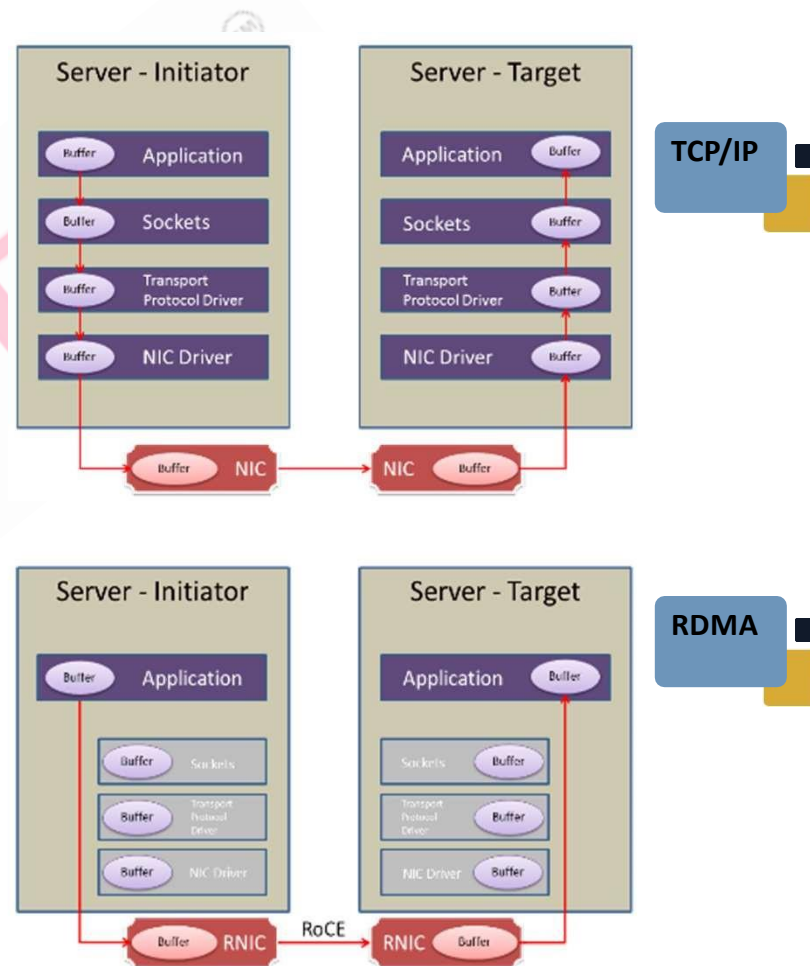


INTEL[®]



RDMA in SuperComputing Clusters

- Remote Direct Memory Access (RDMA)
- Advance transport protocol (same layer as TCP and UDP)
- Main features
 - Remote memory read/write semantics in addition to send/receive
 - Kernel bypass / direct user space access
 - Full hardware offload for network stack
 - Secure, channel based IO
- Application Advantage
 - Lowest latency
 - Highest bandwidth
 - Lowest CPU consumption
- Verbs: RDMA SW Interface (Equivalent to Sockets)





GPUDirect RDMA

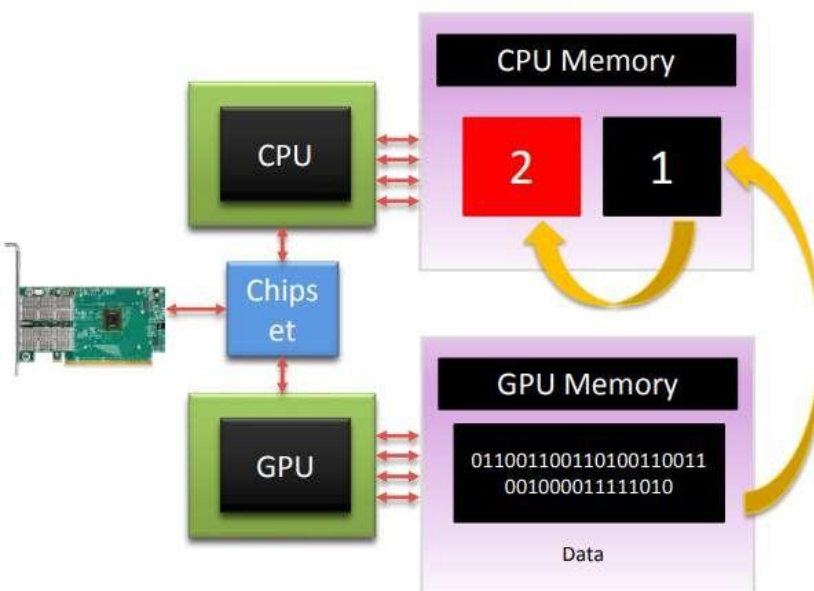
■ Prior to GPUDirect

- GPU use driver-allocated pinned memory buffer
- RDMA use pinned buffers for zero-copy kernel-bypass communication

■ Impossible for RDMA drivers to pin memory allocated by GPU

■ Two copies

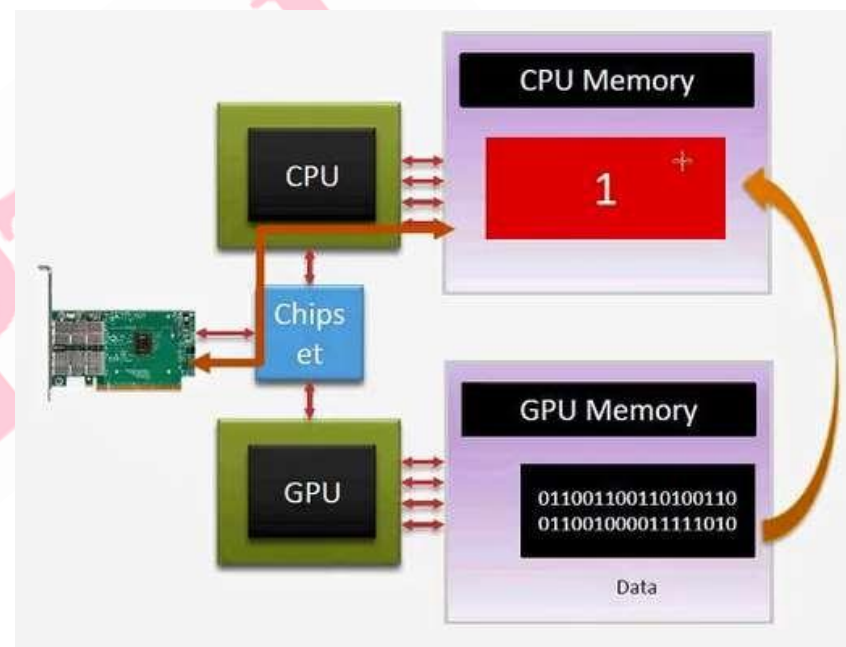
- GPU copies data from GPU internal memory to GPU driver system pinned memory (1)
- User space needs to copy data between the GPU driver system pinned memory (1) and RDMA system pinned memory (2)
- RDMA device sends data to network





GPUDirect RDMA Evolution (cont.)

- **GPUDirect / GPUDirect P2P (Peer-to-Peer)**
 - GPU and RDMA devices share the same pinned memory buffer
- One copy
 - GPU copies data from GPU internal memory to system pinned memory (1)
 - RDMA device sends data to network





GPUDirect RDMA Evolution (cont.)

■ GPUDirect RDMA

- GPU memory is exposed to RDMA NIC
- Direct data path from GPU to network - Data path is zero copy
- **CPU** is involved in the control path - WQE preparation, ring doorbell, handles completions for incoming packets to GPU

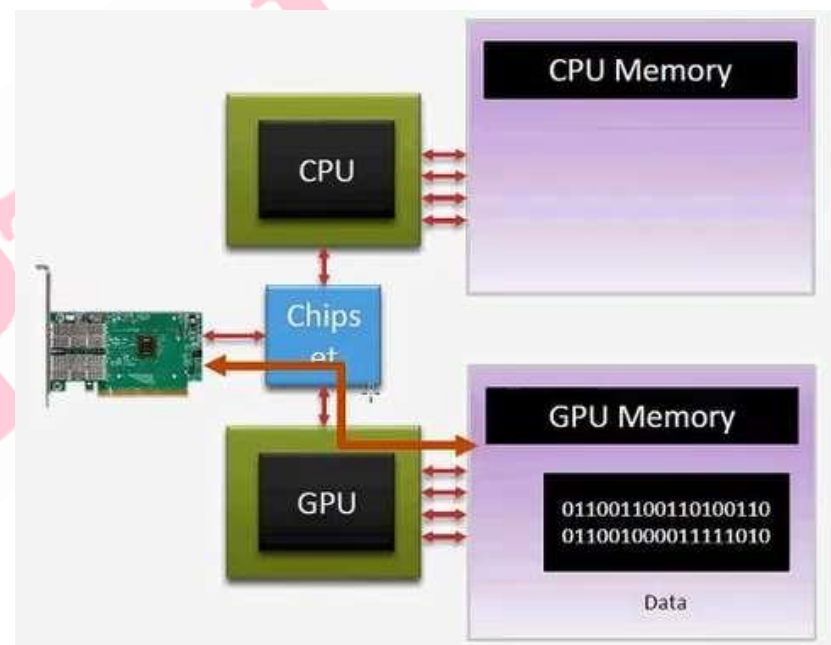
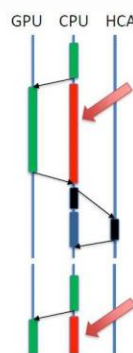
■ Zero copy

- RDMA device sends data to network from GPU memory
- RDMA device receive data from network to GPU memory

■ The CPU still synchronizes between GPU tasks and data transfers

```
while(fin) {  
    gpu_kernel <<<... , stream>>>(buf);  
    cudaStreamSynchronize(stream);  
    ibv_post_send(buf);  
    ibv_poll_cq(cqe);  
}
```

100% CPU Utilization





GPUDirect RDMA Evolution (cont.)

■ GPUDirect RDMA Async

- GPU memory is exposed to RDMA NIC
- Direct data path from GPU to network - Data path is zero copy
- **CPU** is involved in WQE preparation and release completed WQEs
- **GPU** is involved in Ring Doorbell, Handles completions for incoming packets to GPU

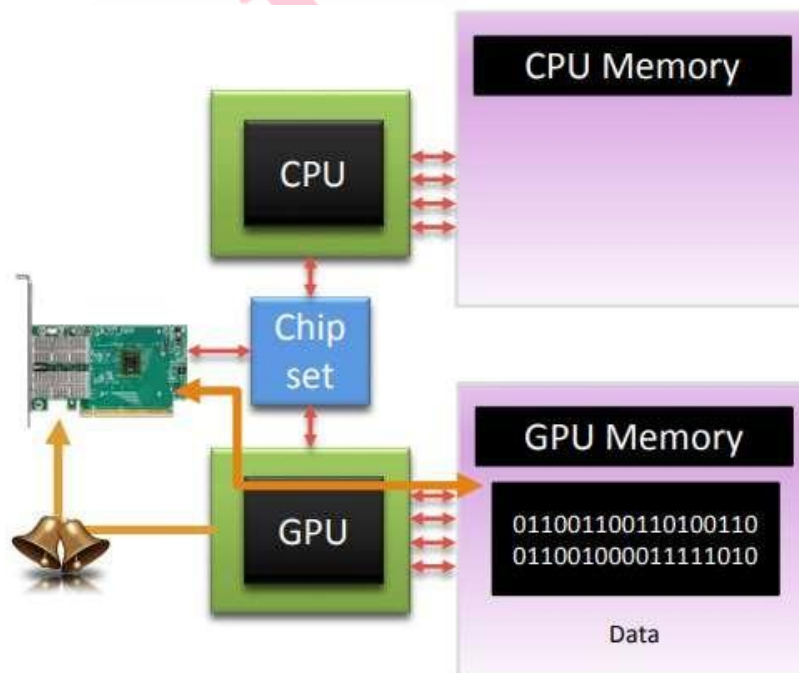
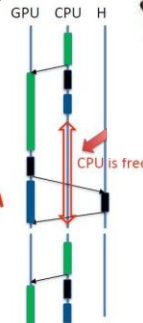
■ Zero copy

- RDMA device sends data to network from GPU memory
- RDMA device receive data from network to GPU memory

■ Reduce CPU utilization

```
while(fin) {  
    gpu_kernel <<<... , stream>>>(buf);  
    gds_stream_queue_send(stream, qp, buf);  
    gds_stream_wait_cq(stream, cq);  
}
```

No CPU in critical path



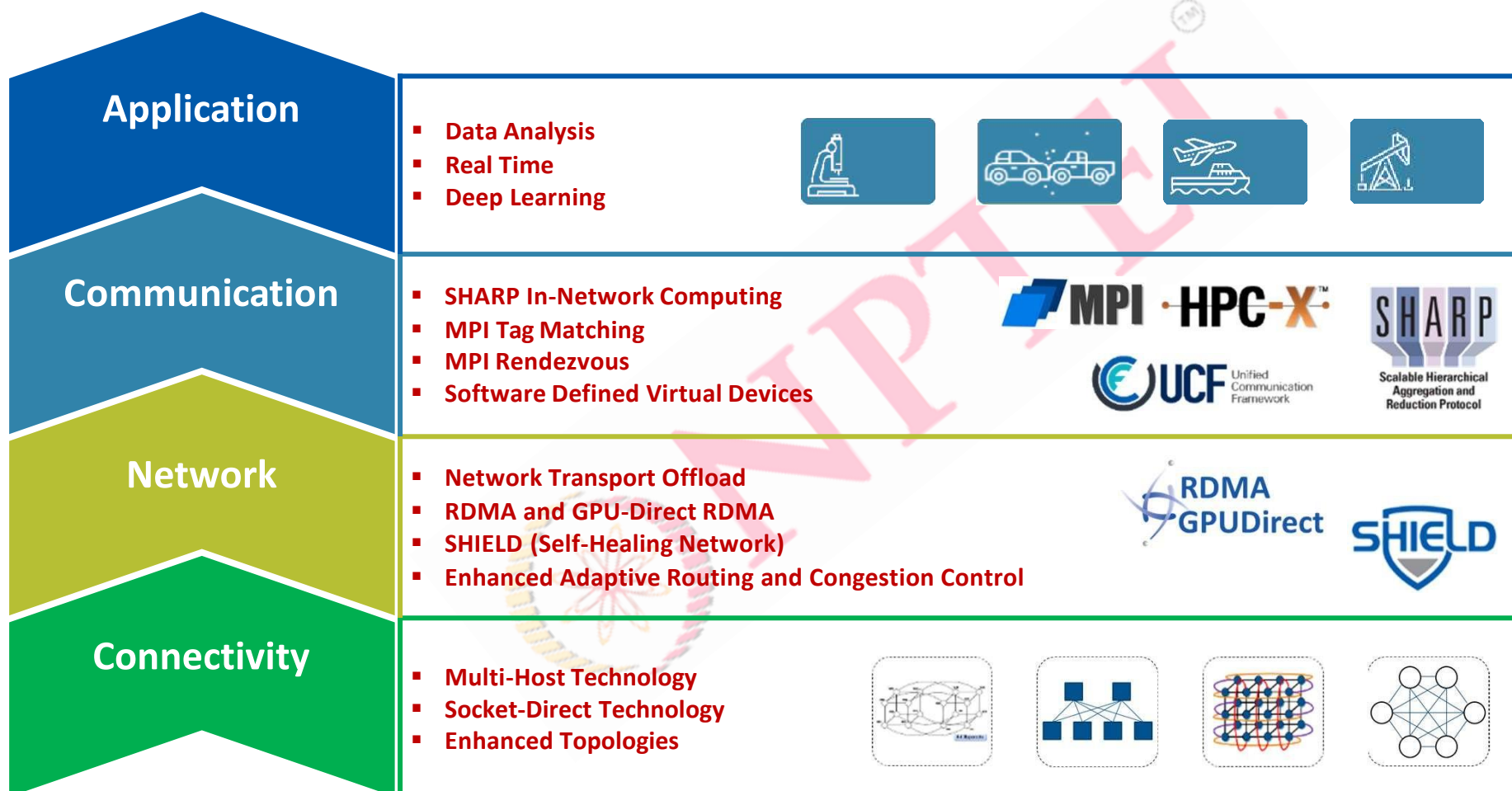


Latency Contribution of Network Components

	Copper Cables	5ns per m					
	Optical Cables	5ns per m + 20ns*		5ns per m + 100ns			
	40p 1U Switch		~100ns	-400ns			-2000ns
	800p Chassis Switch			<350ns			-9000ns
	ConnectX-6 B2B		650ns				7000ns
	MPI OSU latency		~900ns				-10000ns



Need to Accelerate All Levels of HPC / AI Frameworks



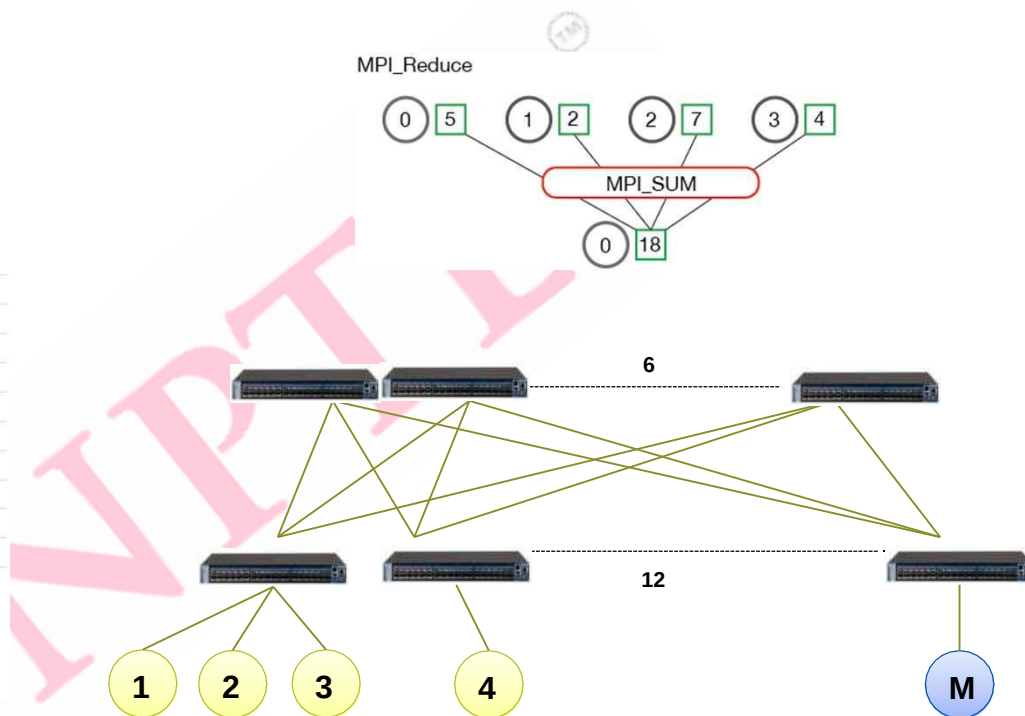
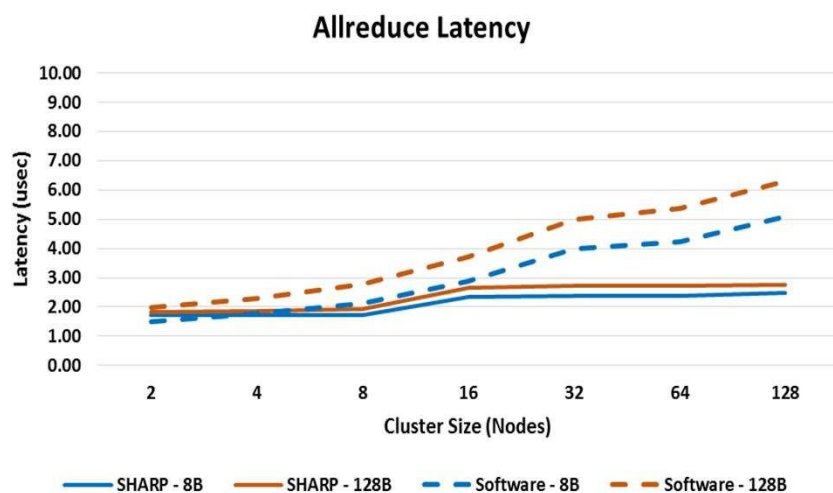


Advance Technologies and “Intelligent” Networks





Collectives like MPI all_reduce

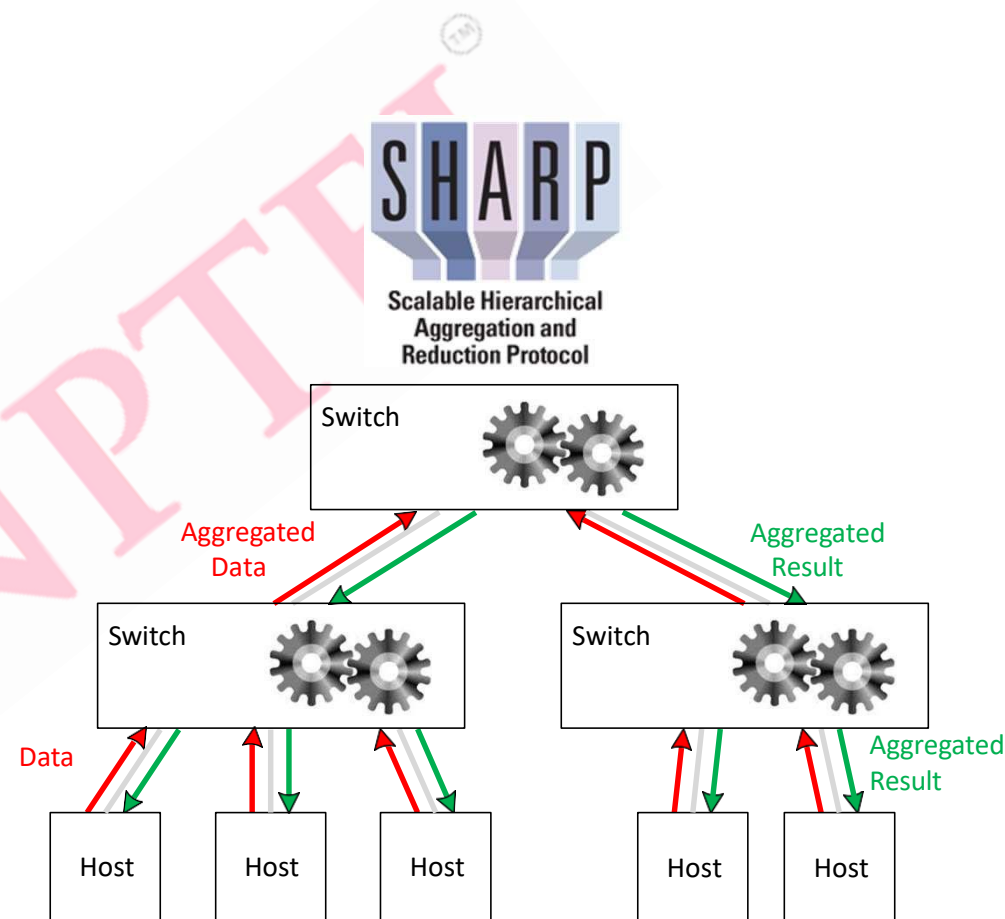


**SHARP enables 75% Reduction in Latency
Providing Scalable Flat Latency**



Scalable Hierarchical Aggregation and Reduction Protocol (SHARP)

- Reliable Scalable General Purpose Primitive
 - In-network Tree based aggregation mechanism
 - Large number of groups
 - Multiple simultaneous outstanding operations
- Applicable to Multiple Use-cases
 - HPC Applications using MPI / SHMEM
 - Distributed Machine Learning applications
- Scalable High Performance Collective Offload
 - Barrier, Reduce, All-Reduce, Broadcast and more
 - Sum, Min, Max, Min-loc, max-loc, OR, XOR, AND
 - Integer and Floating-Point, 16/32/64 bits

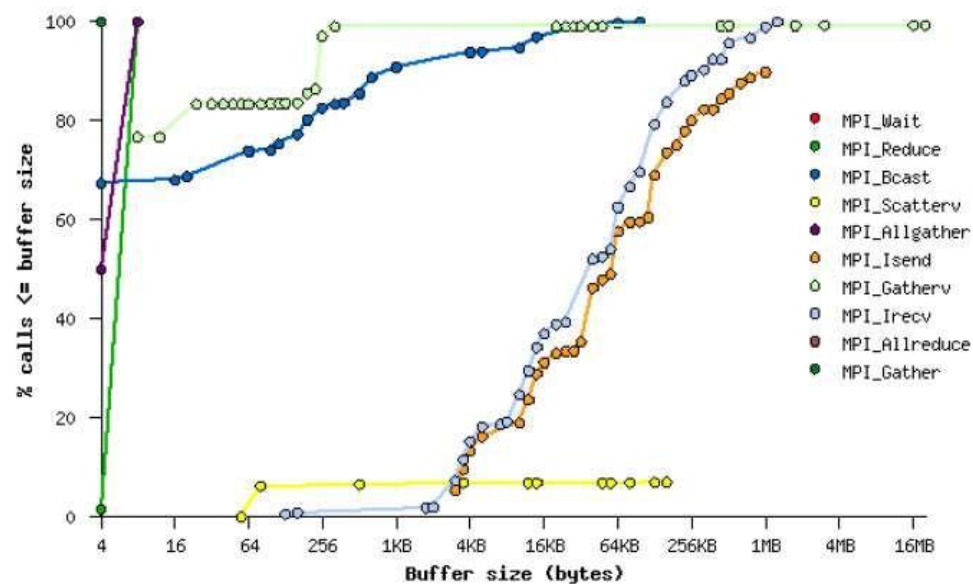
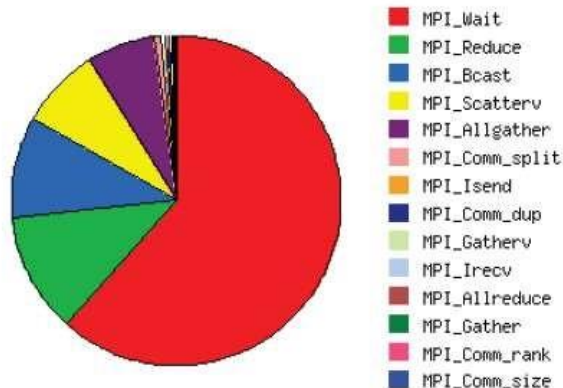




MPI profile of an application (eg. WRF)

Communication

% of MPI Time

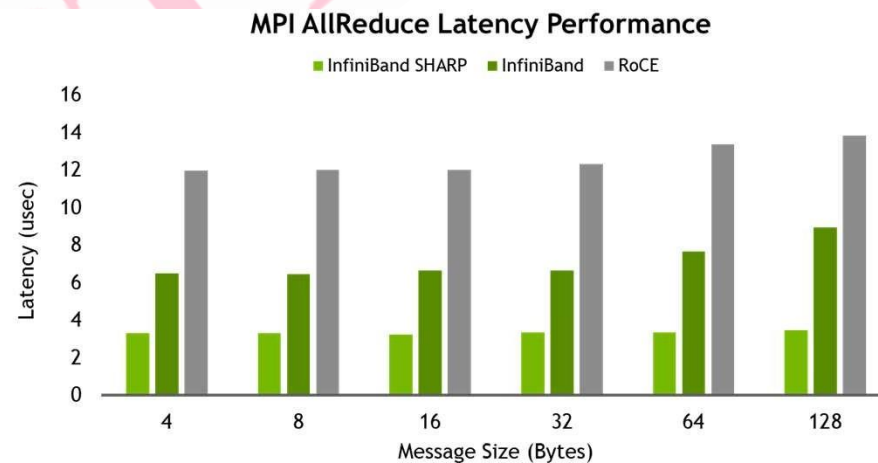
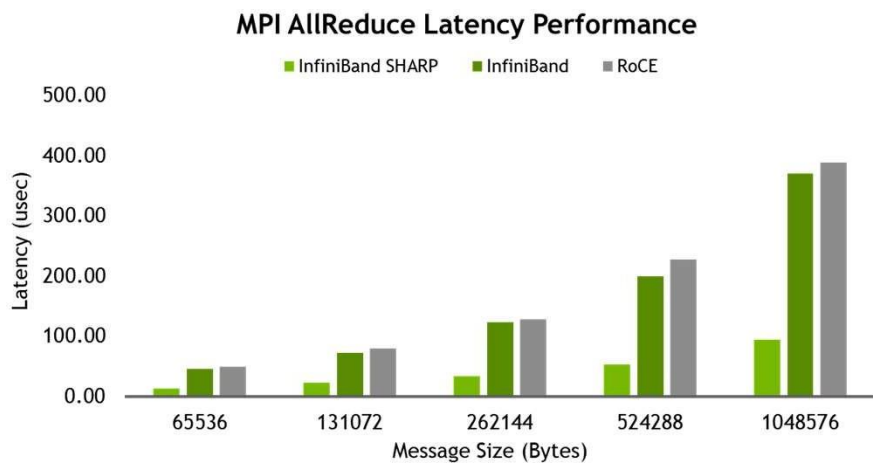


- Take MPI_Reduce for example
- In this application for eg; 4-8 byte MPI_Reduce has a significant component
- SHARP can reduce this latency from ~100us to <10us for a 512 node run



InfiniBand SHARP Performance Advantage

4X Higher Performance





SHARP Accelerates AI Performance

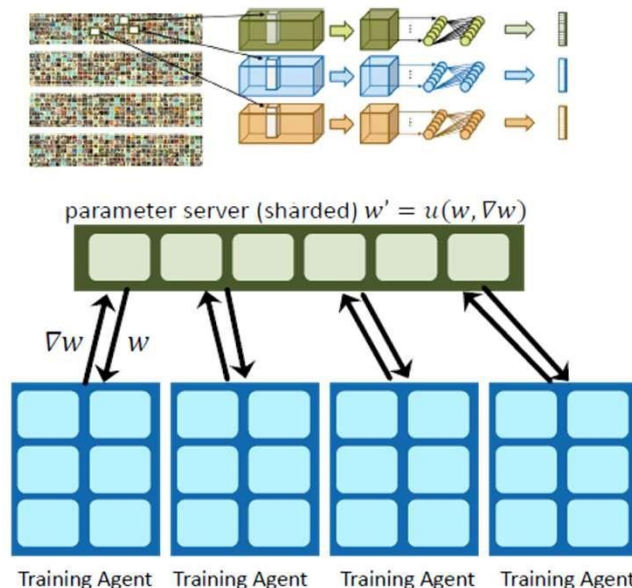
The CPU in a parameter server becomes the bottleneck



Scalable Hierarchical
Aggregation and
Reduction Protocol



Performs the Gradient Averaging
Replaces all physical parameter servers
Accelerate AI Performance





Nvidia A100 GPU

	A100 40GB PCIe	A100 80GB PCIe	A100 40GB SXM	A100 80GB SXM
FP64	9.7 TFLOPS			
FP64 Tensor Core	19.5 TFLOPS			
FP32	19.5 TFLOPS			
Tensor Float 32 (TF32)	156 TFLOPS 312 TFLOPS*			
BFLOAT16 Tensor Core	312 TFLOPS 624 TFLOPS*			
FP16 Tensor Core	312 TFLOPS 624 TFLOPS*			
INT8 Tensor Core	624 TOPS 1248 TOPS*			
GPU Memory	40GB HBM2	80GB HBM2e	40GB HBM2	80GB HBM2e
GPU Memory Bandwidth	1,555GB/s	1,935GB/s	1,555GB/s	2,039GB/s
Max Thermal Design Power (TDP)	250W	300W	400W	400W
Multi-Instance GPU	Up to 7 MIGs @ 5GB	Up to 7 MIGs @ 10GB	Up to 7 MIGs @ 5GB	Up to 7 MIGs @ 10GB
Form Factor	PCIe		SXM	
Interconnect	NVIDIA® NVLink® Bridge for 2 GPUs: 600GB/s ** PCIe Gen4: 64GB/s		NVLink: 600GB/s PCIe Gen4: 64GB/s	
Server Options	Partner and NVIDIA-Certified Systems™ with 1-8 GPUs		NVIDIA HGX™ A100-Partner and NVIDIA-Certified Systems with 4,8, or 16 GPUs NVIDIA DGX™ A100 with 8 GPUs	



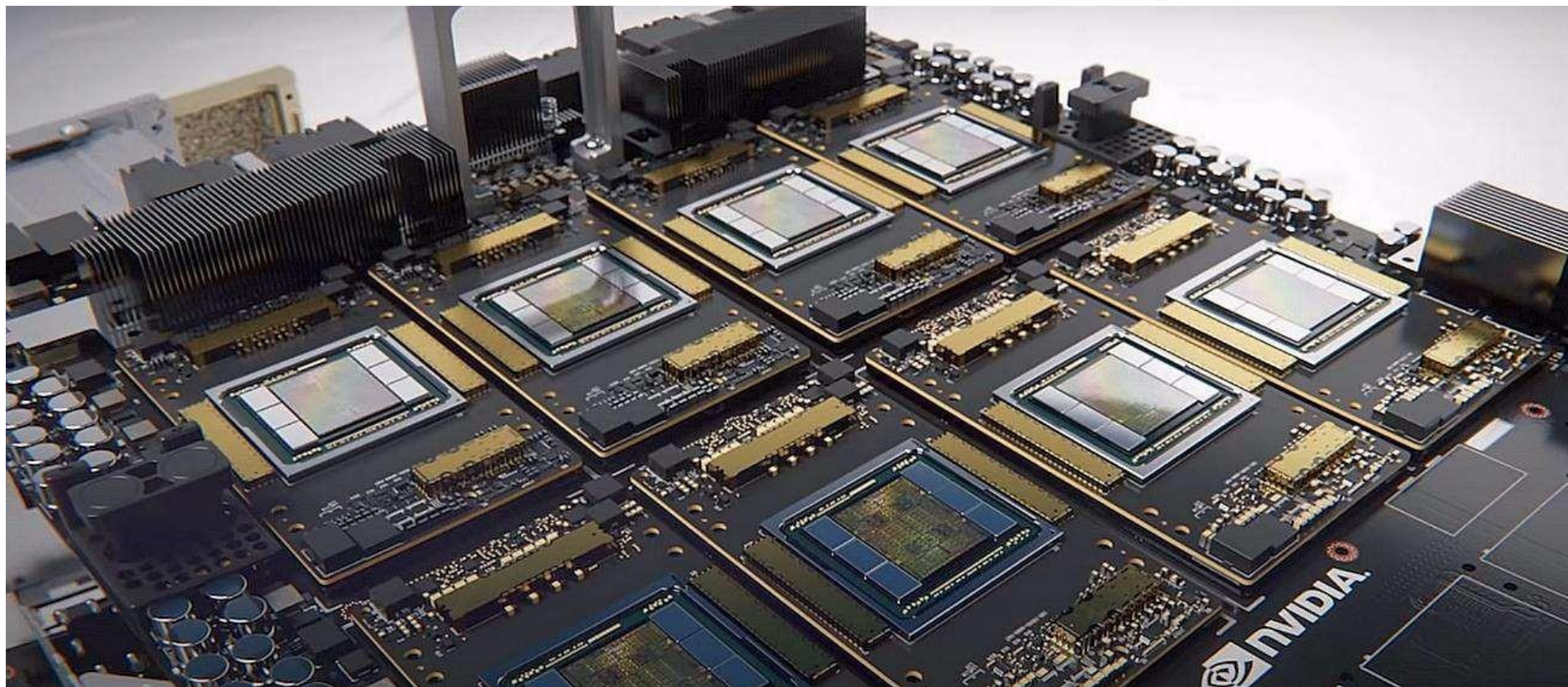
A100 PCIe



A100 SXM

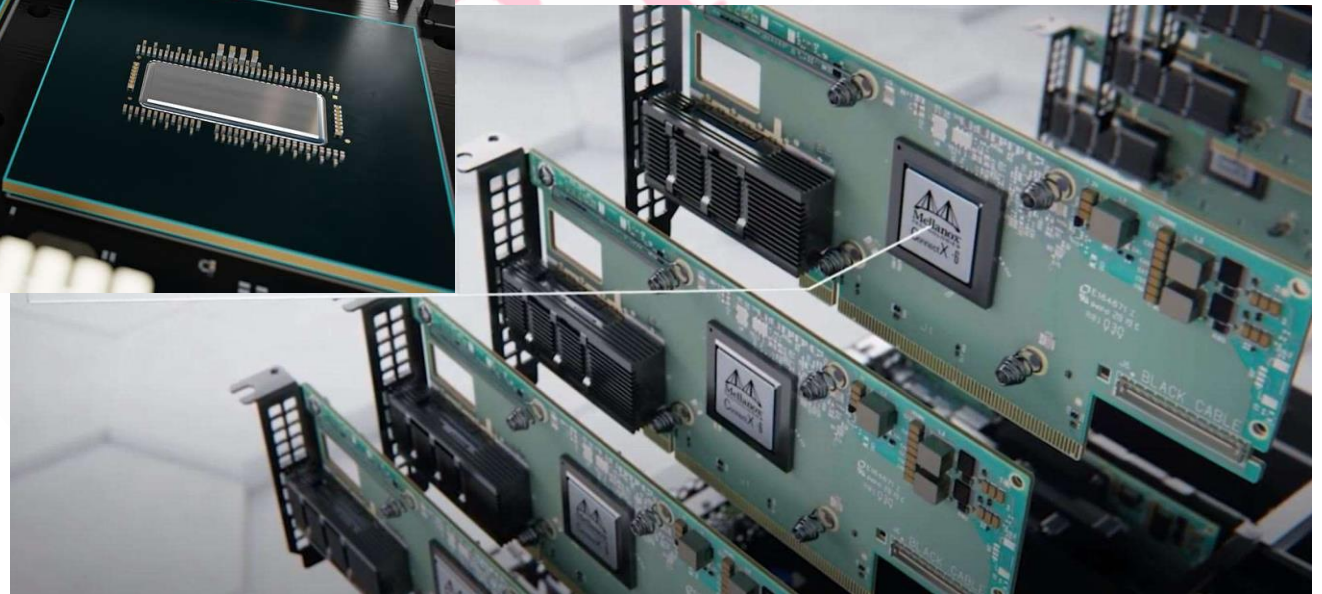
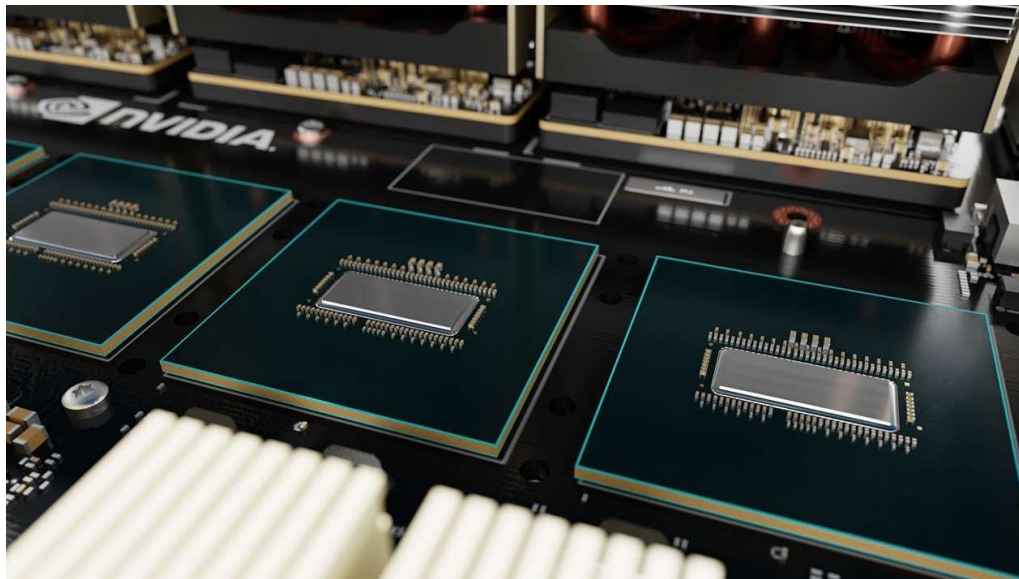


8x A100 GPUs





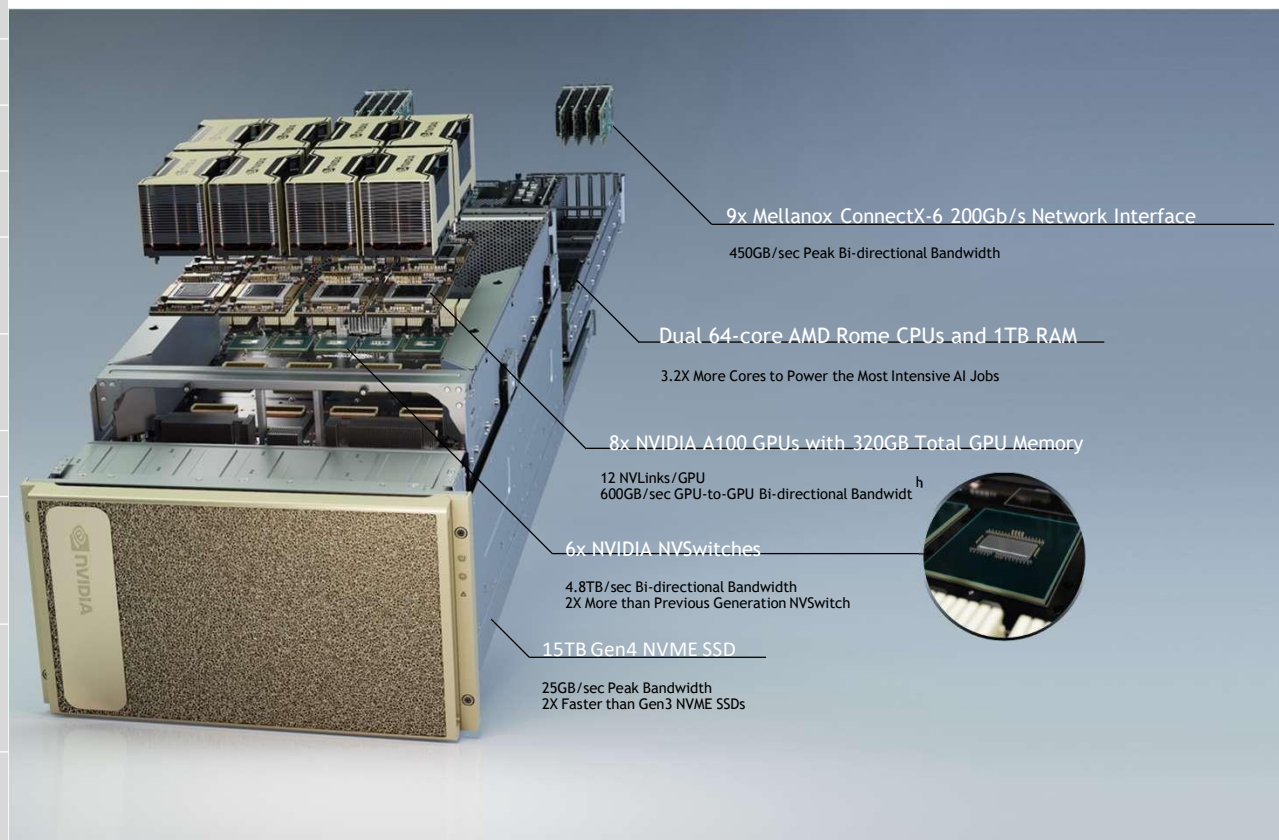
Internal and External Networking





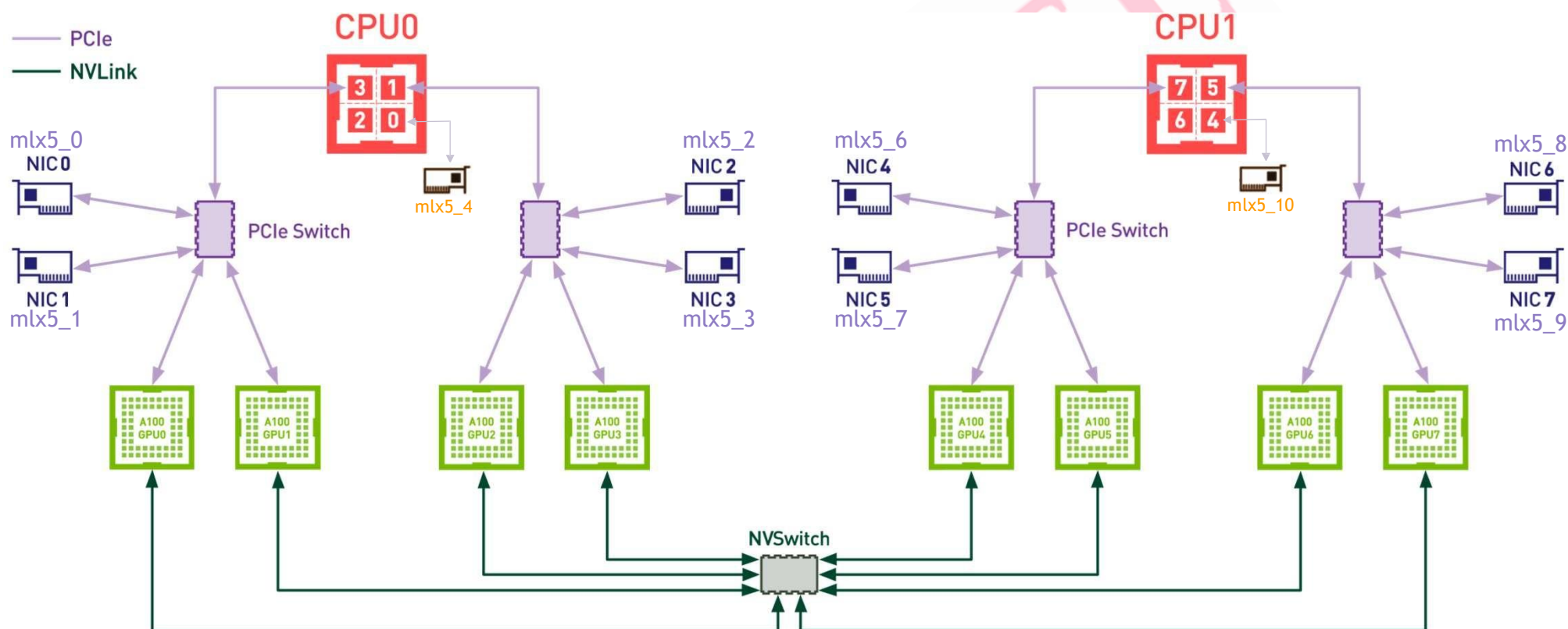
Game-changing performance for innovators

App Focus Components	
GPUs	8x NVIDIA A100 Tensor Core GPUs
GPU Memory	320GB Total
NVIDIA NVSwitch	6
Performance	5 petaFLOPS AI 10 petaOPS, INT8
CPU	Dual AMD Rome, 128 cores total, 2.25 GHz (base), 3.4 GHz (max boost)
System Memory	1TB
Networking	9x Mellanox ConnectX-6 VPI HDR InfiniBand/200GigE 10 th Dual-port ConnectX-6 optional
Storage	OS: 2x 1.92TB M.2 NVME drives Internal Storage: 15TB (4x 3.84TB) U.2 NVME drives
System Power Usage	6.5 kW Max
System Weight	271 lbs (123 kgs)



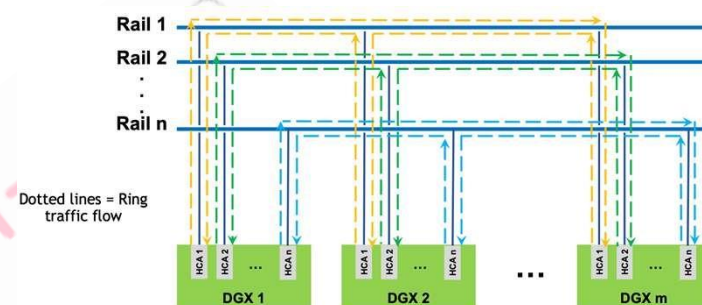
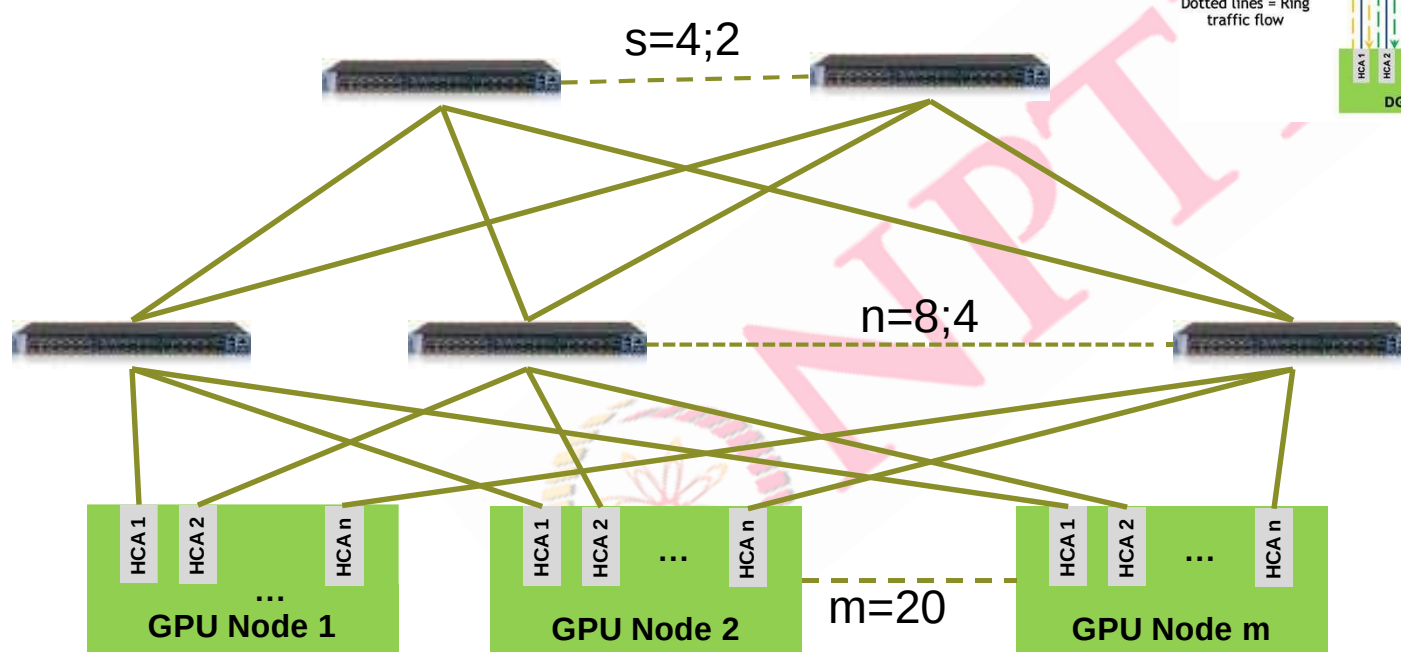


Typical GPU accelerated server - DGX A100





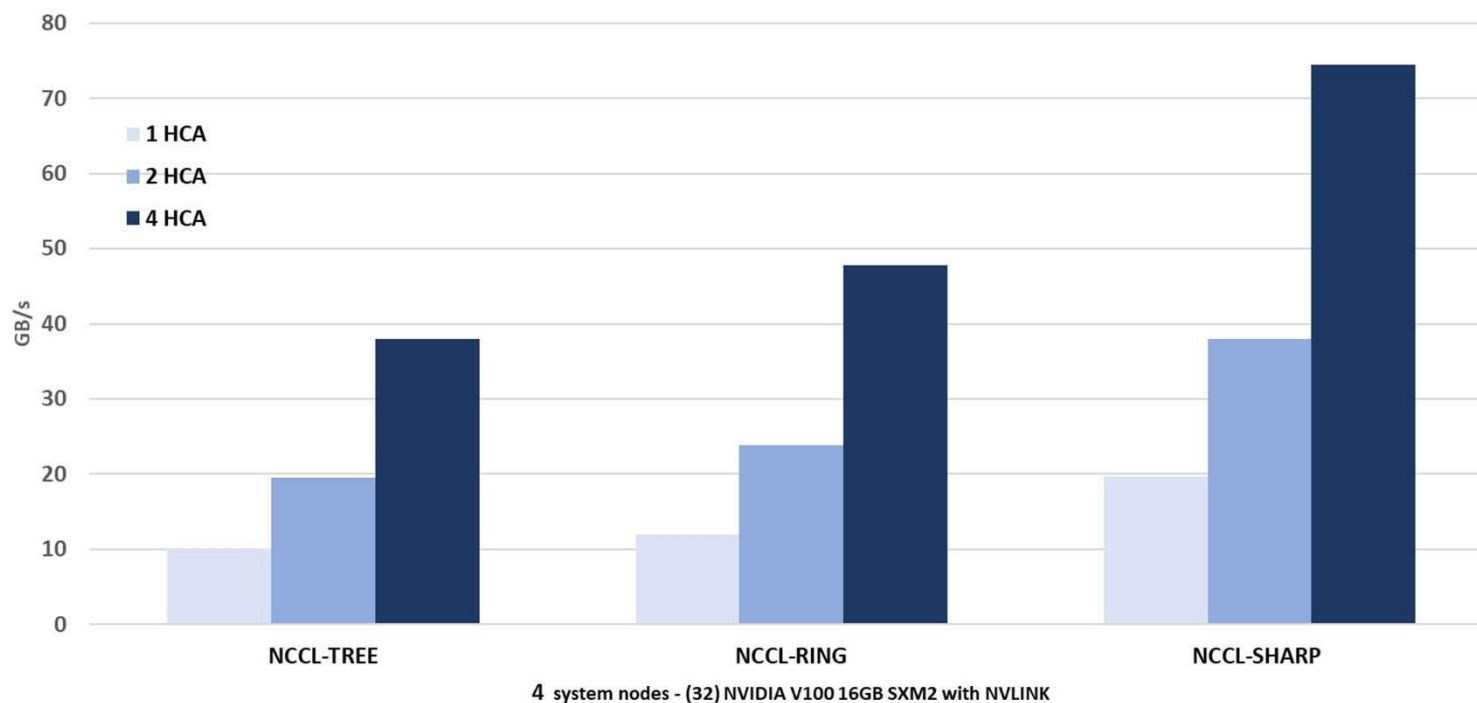
Mapping Rails To IB Trees





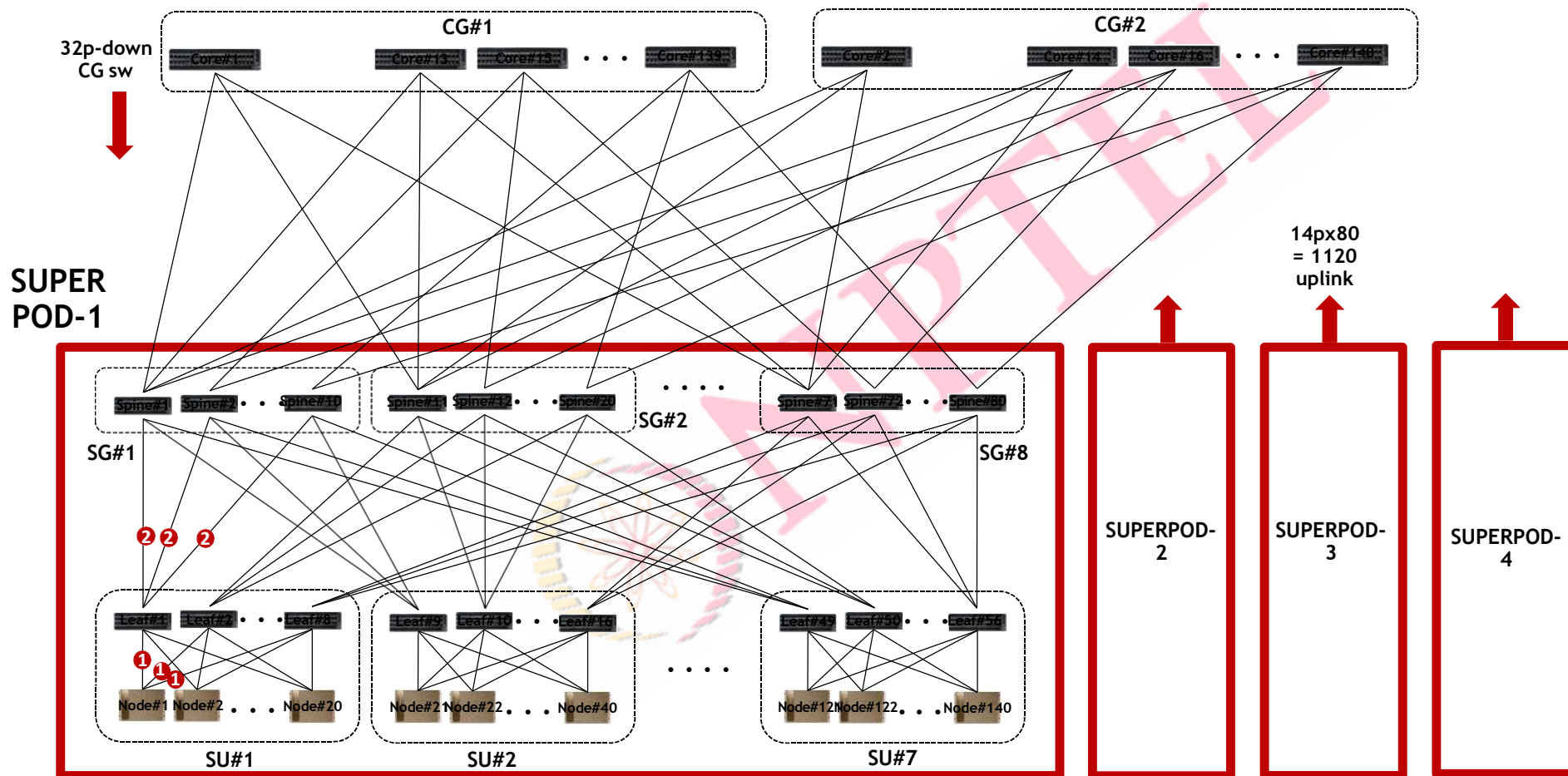
SHARP Delivers Highest Performance for AI

Mellanox SHARP Plug-in for NCCL 2.4
(Bandwidth)





4x SuperPod network topology





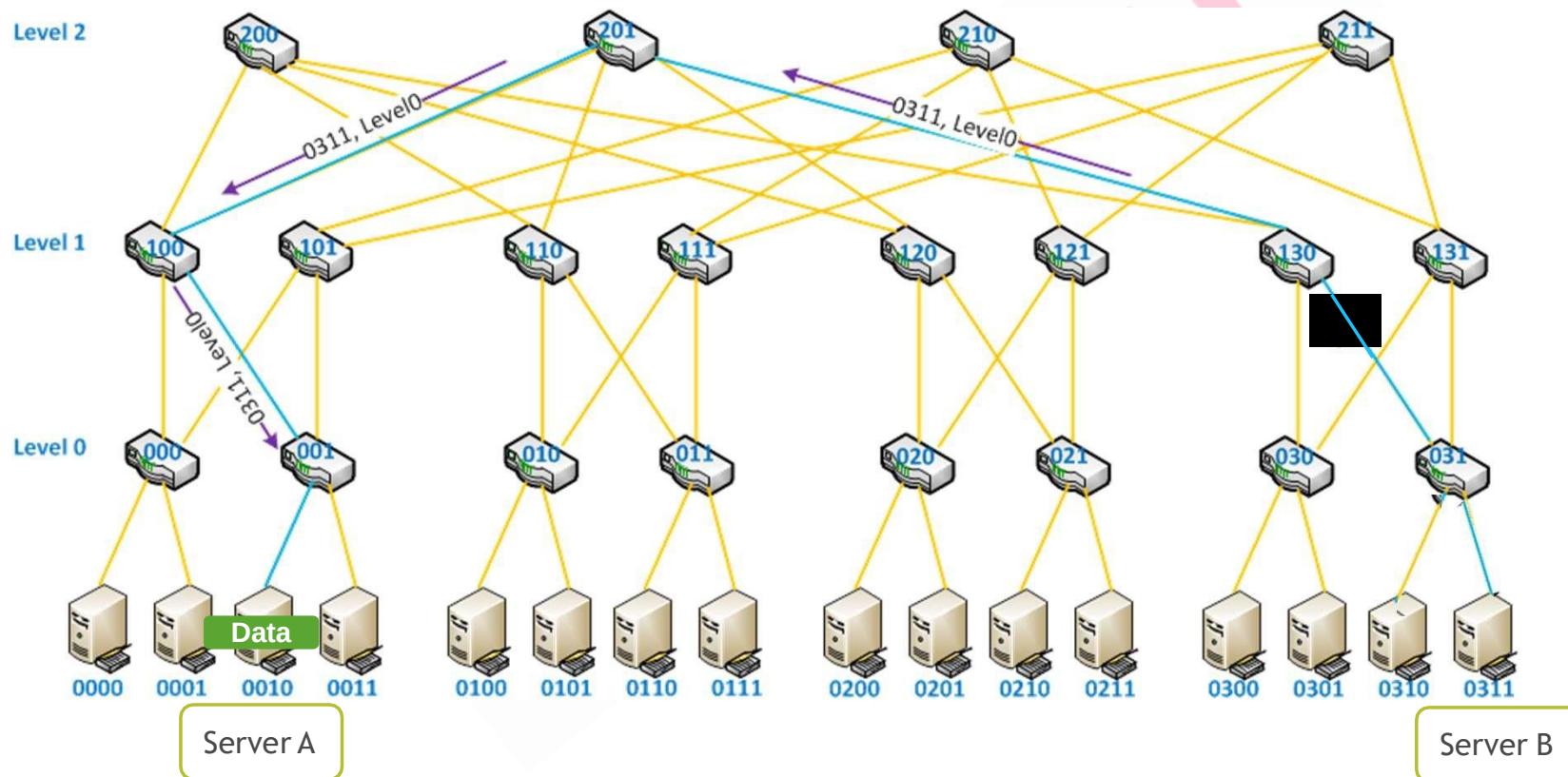
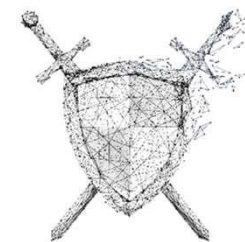
Resilient and “Self Healing” Networks



NIPTEELTM

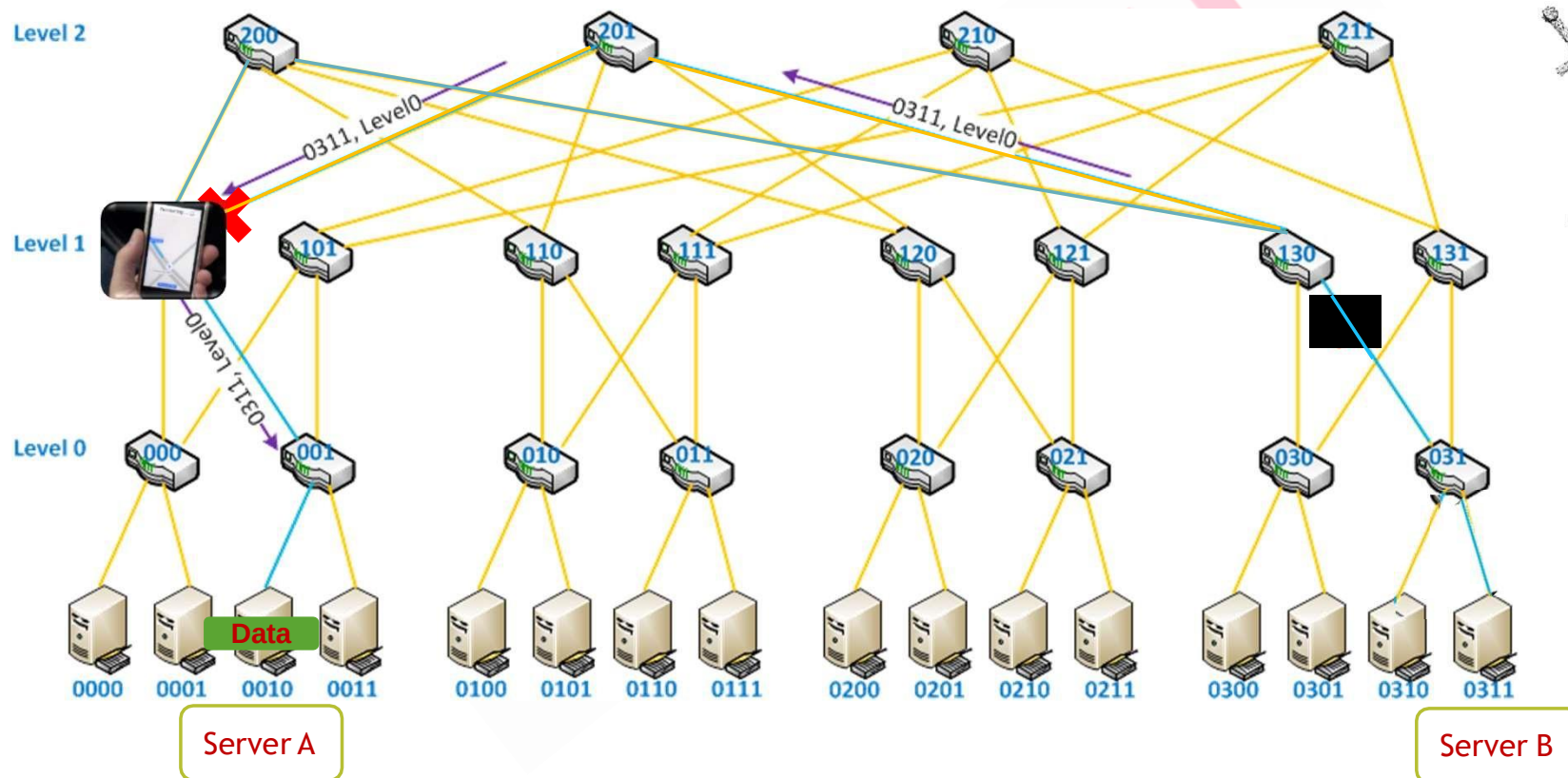


Example: Consider a Flow From A to B



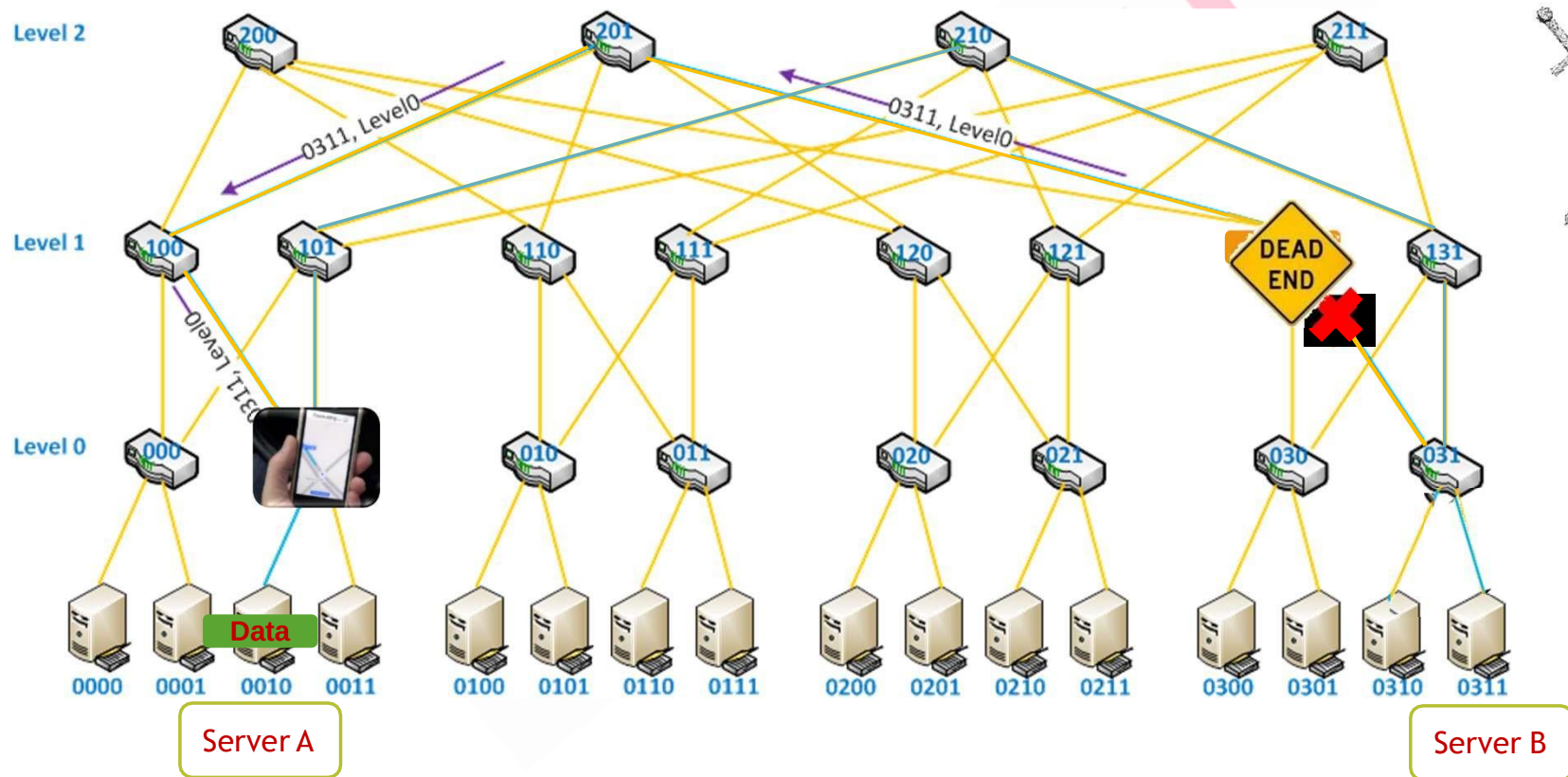


Example: The Simple Case: Local Fix



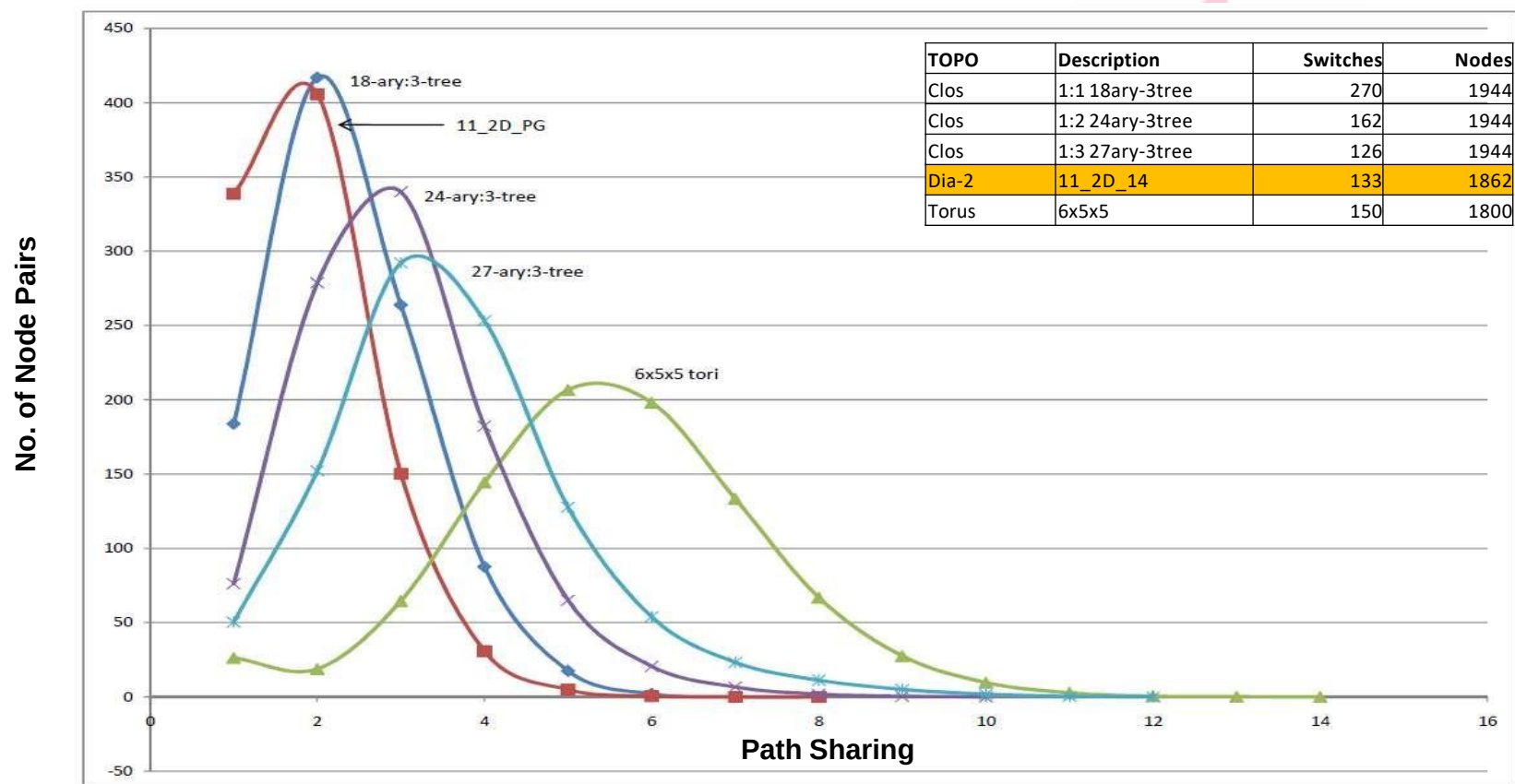


Example: The Remote Case - Using Fault Recovery Notifications





And Some Simple Results

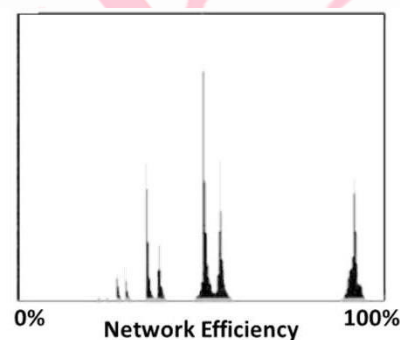




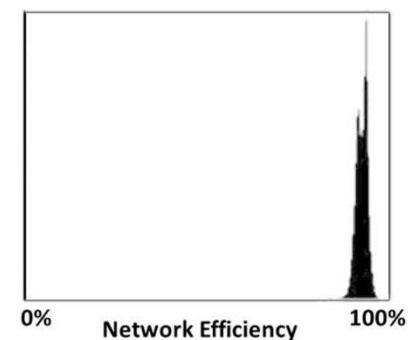
Adaptive Routing (AR) Performance

- Oak Ridge National Laboratory – Coral Summit supercomputer
- Bisection bandwidth benchmark, based on mpiGraph
 - Explores the bandwidth between possible MPI process pairs
- AR results demonstrate an average performance of 96% of the maximum bandwidth measured

mpiGraph explores the bandwidth between possible MPI process pairs. In the histograms, the single cluster with AR indicates that all pairs achieve nearly maximum bandwidth while single-path static routing has nine clusters as congestion limits bandwidth, negatively impacting overall application performance.



Without Adaptive Routing

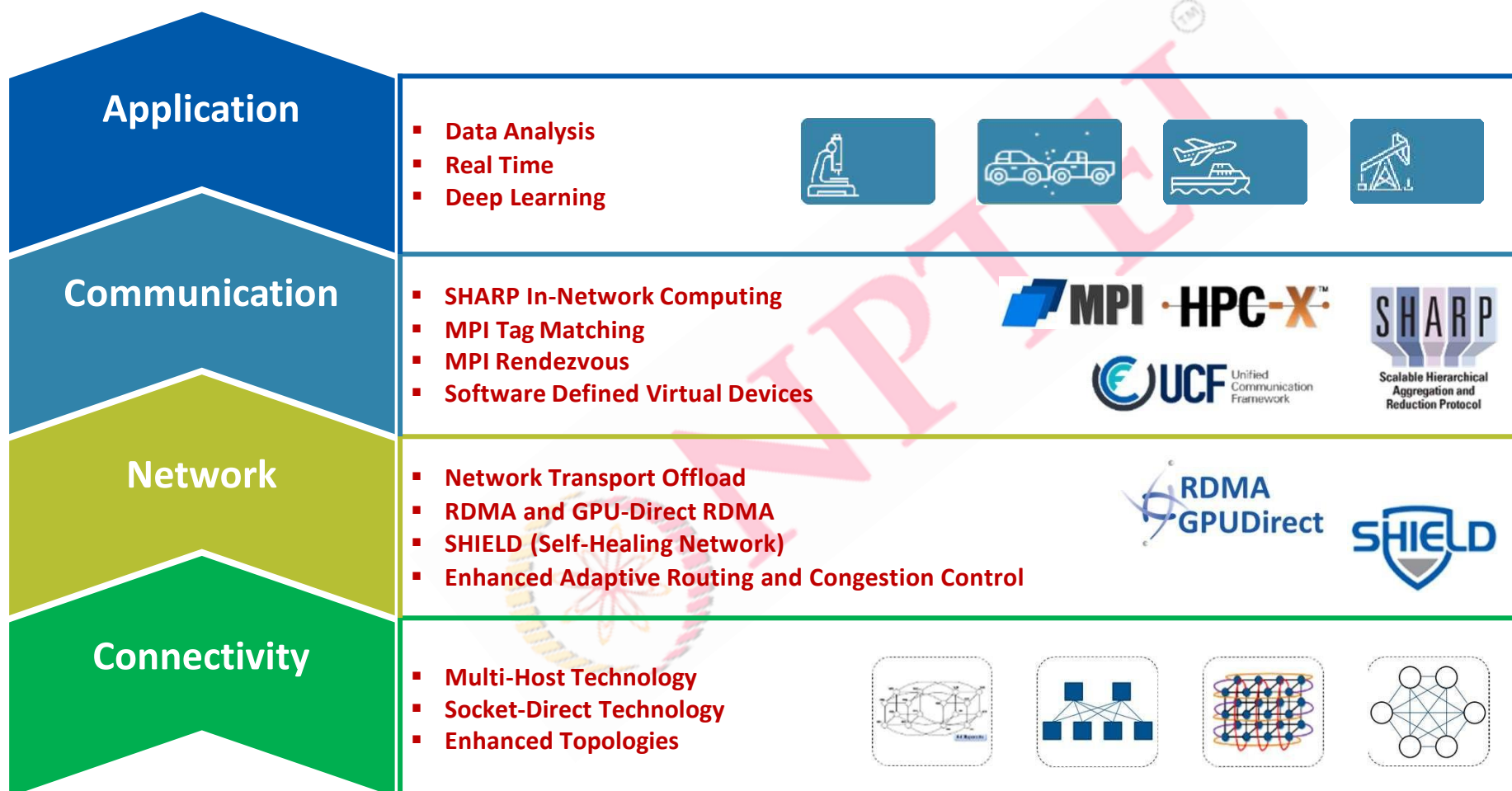


With Adaptive Routing

Summit's MpiGraph Output



Accelerating All Levels of HPC / AI Frameworks

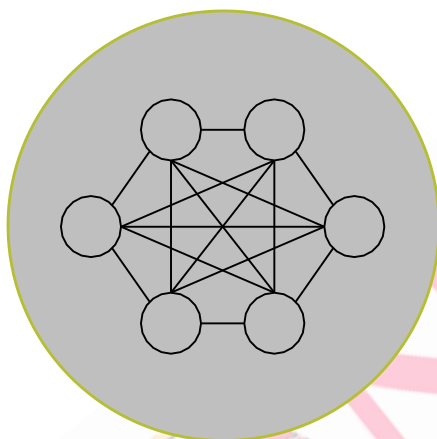




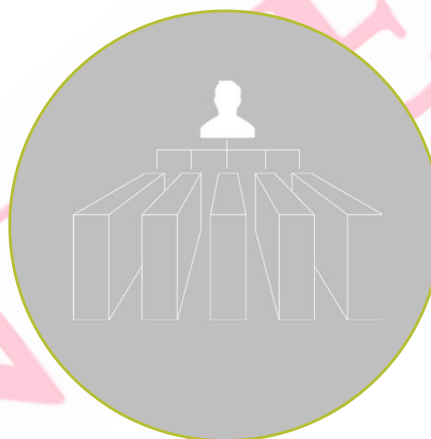
InfiniBand Technology Fundamentals



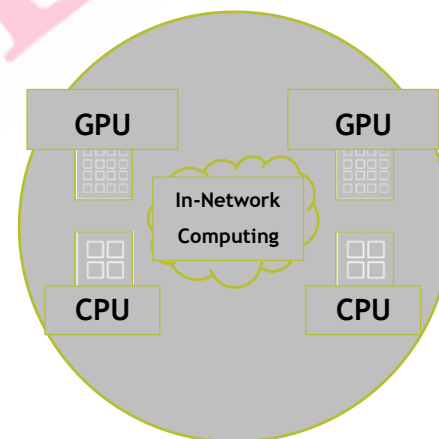
Standard



Architected to Scale



**Software Defined Network
Centralized Management**



In-Network Computing

Thank You



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